Improving Environmental Outcomes Through Investments in Local Governance: Evidence from Brazil*

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Abstract

The Green Municipalities Program (Programa Municípios Verdes–PMV) has operated in the Brazilian state of Pará since 2010. Though voluntary, the program has provided $20M in funding to local governments to combat deforestation in the Brazilian Amazon through improvements to local systems and bureaucracies. By leveraging advanced causal inference methods and twenty years of municipality and property-level land use data, we examine the program's effects on deforestation, secondary forest regeneration, and local governance outcomes. We find that PMV is associated with a 12% annual reduction in deforestation and a 7% increase in secondary forest cover, though deforestation results are sensitive to the inclusion of municipalities previously subject to federal blacklisting. Secondary forest recovery appears more robust and concentrated in younger forests and medium-sized properties, suggesting improved land use incentives and enforcement capacity. We also find evidence of reduced land conflict and increased non-agricultural economic activity. Under conservative carbon valuation assumptions, estimated climate benefits exceed program costs. These findings point to the potential of subnational governance investments to support conservation and sustainable development objectives in tropical forest regions.

Keywords: Environmental policy, local governance, deforestation, secondary forest growth

JEL classification: O2, Q18, Q23, Q28

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1 Introduction

Forests represent one of our planet's most critical natural defenses against climate change, yet they remain under persistent threat worldwide. The dual challenges of deforestation and forest degradation not only accelerate carbon emissions but also erode biodiversity and threaten millions who depend on forest ecosystems for their livelihoods. While command-and-control approaches like strict regulations and penalties have shown some success in reducing forest loss, their effectiveness often comes at significant social and economic costs to local communities. An alternative and potentially complementary approach focuses on strengthening local governance systems and providing positive incentives for conservation. Such "carrot" rather than "stick" approaches may not only help prevent deforestation but could also create conditions favorable for secondary forest regeneration—the natural or assisted regrowth of forests on previously cleared land—which provides crucial unaccounted for carbon sequestration benefits. These governance-focused initiatives merit closer examination, particularly in regions like the Brazilian Amazon where centralized enforcement faces significant implementation challenges across vast territories, and where secondary forest recovery represents an underappreciated opportunity for climate mitigation.

This study investigates the effects of the Green Municipality Program (Programa Municípios Verdes–PMV), a voluntary conservation initiative for local governments in the Brazilian Amazon. We study how the PMV program has influenced deforestation rates, the recovery of secondary forests, and other local economic and governance outcomes.

Born out of the desire for an alternative to strict command and control conservation efforts, the PMV program was launched by the Brazilian state of Pará in 2010 and funded by the Amazon Fund. Prior work has studied the effects of the program on deforestation (Sills et al. 2020; Moz-Christofoletti et al. 2022). We re-examine the effects of PMV on deforestation with new data and new methods. Further, as a novel contribution, we also test whether the PMV improved environmental outcomes beyond its main intended target of curtailing deforestation. Specifically, we test whether PMV, through strengthening local

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capacity, led to increased forest recovery (secondary forest growth). We then delve deep into the mechanisms producing these outcomes, focusing on how PMV changed local bureaucratic systems and capacity.

Environmental federalism theory suggests decentralized governance can improve resource management efficiency by better matching policies to local conditions and preferences (Oates 1972; Tiebout 1956). In practice, however, the effect of decentralization on environmental outcomes is critically dependent on local institutional capacity (Ostrom 2010). Local governments often face significant resource constraints and technical limitations that can impede effective environmental management (Andersson et al. 2007). This problem is particularly acute in developing regions where budget shortfalls and limited administrative capacity coincide with high deforestation pressures (Burgess et al. 2012; Alix-Garcia et al. 2016). Additionally, political economy considerations suggest decentralization may exacerbate resource misallocation when local elites capture decision-making processes, potentially undermining conservation goals (Bardhan et al. 2000). These challenges highlight why decentralization policies that simultaneously strengthen local governance capacity may outperform simple authority transfers. When decentralization includes investments in monitoring systems, enforcement capabilities, and administrative capacity, theory predicts more effective environmental protection through reduced information asymmetries and transaction costs (Laffont et al. 1993; Besley et al. 2007).

PMV represents an ideal empirical setting to evaluate these theoretical predictions, as it combines an attempt to decentralized environmental management with targeted capacity-building investments. The program extends beyond traditional command-and-control approaches by fostering improvements in local governance and land regularization[1], while engaging local stakeholders through "townhall" style meetings.

[1]In the Brazilian context, land regularization refers to the formal legal process of granting property rights to individuals or communities who occupy land without formal titles or legal ownership, often in rural areas and informal urban settlements. This process is intended to improve security of tenure, promote sustainable land use, and protect vulnerable communities, while aligning land use with environmental and social regulations. Please see, Alston et al. 2011; Sparovek et al. 2019.

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We rely on both satellite and administrative data in order to test the effects of PMV. We collect deforestation data from two different sources: MapBiomas, an independent organization, and PRODES, the official dataset from the Brazilian government. These datasets provide a comprehensive measure of deforestation using satellite data and ground truth information, enabling us to gain a deeper understanding of land use change and primary forest clearing. We also utilize novel high-resolution satellite data to identify secondary forest growth and determine the age of these forests (Junior et al. 2020). Further, we rely on a property-level dataset to ascertain whether the impacts of the policy are driven by small, medium, or large landholders. Finally, we analyze data on land-related conflicts, forest fires, and non-agricultural economic production to explore the policy's effects on a diverse set of environmental and socio-economic outcomes.

To identify the impact of PMV, we employ generalized synthetic differences-in-differences (Xu 2017a) to address the challenge of non-random selection of municipalities into the program. This approach allows us to establish credible counterfactuals in our setting with voluntary, staggered adoption. In supplementary analysis, we confirm that our findings remain consistent across alternative specifications, including two-way fixed effects regressions, staggered difference-in-differences models (Callaway et al. 2021), and interactive fixed effects approaches (Liu et al. 2021), reinforcing the robustness of our results. However, given the voluntary nature of program uptake and its overlap with concurrent command-and-control policies—particularly Brazil's "blacklisting" program, which imposed strict penalties on high-deforesting municipalities and helped motivate PMV's alternative, governance-based approach—we interpret our results as plausibly reflecting PMV's influence within a complex policy environment. To help untangle these effects, we extend our analysis to a restricted sample of municipalities that were never blacklisted.

When considering the full sample of municipalities, we find that participation in PMV is associated with reductions in deforestation and increases in secondary forest cover. PMV led to a 12% reduction in deforestation, corresponding to a total decrease of 5,684 km^2

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in the treated group of municipalities. Moreover, these areas experienced a substantial total increase of 36,663 km^2 in secondary forest extent, representing a 7% rise compared to non-PMV municipalities. To put this into perspective, 5,684 km^2 is roughly equivalent to an area the size of Delaware, while 36,663 km^2 is comparable to the size of Taiwan. When restricting the analysis to municipalities never blacklisted, the deforestation effect becomes mixed and statistically insignificant in our preferred MapBiomas specification, but the increase in secondary forest cover remains robust.

Additionally, our results show that most of the increase in secondary forests was driven by young secondary forest growth, which grew by 11% within participating municipalities. By exploiting data on all registered rural properties in Brazil, we are able to document that a large fraction of these effects are coming from registered rural properties, and are mostly driven by medium-sized properties. These findings confirm our expectations that improved land regularization should lead to additional land use benefits on top of deforestation reduction. We also show that PMV led to important reductions in land related conflict and an increase in non-agricultural economic activity within participating municipalities. By contrast, we show that blacklisting resulted in more environmental fines, had no effect on land conflict, and reduced non-agricultural economic activity—indicating that top-down enforcement may generate different dynamics than locally embedded governance programs.

Finally, we translate the program's environmental impacts into economic terms by applying a range of social cost of carbon (SCC) values to our estimates of avoided deforestation and secondary forest recovery. Using biome-specific carbon coefficients, we estimate that between 2010 and 2019, avoided deforestation under PMV prevented 162.36 million metric tons of avoided CO_2 emissions (i.e., $MtCO_2e$), and secondary forest regrowth removed an additional 18.35 $MtCO_2e$ from the atmosphere. These figures represent approximately 7.5% and 0.85% of Brazil's total greenhouse gas emissions in 2020, respectively.[2] Valuing these impacts at $50 per ton CO_2e yields a net present value (NPV) of $6.27 billion from

[2]Brazil's total greenhouse gas emissions in 2020 were approximately 2.16 billion tonnes of CO_2 equivalent ($GtCO_2e$) or 2,160 $MtCO_2e$ according to SEEG data (De Azevedo et al. 2018).

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avoided emissions and $0.71 billion from forest recovery. At higher SCC values, such as the $185 per ton recently recommended in the literature, the total benefits increase to $23.19 billion and $2.64 billion, respectively. Even under the conservative scenario that excludes blacklisted municipalities and considers only effects from forest regeneration, we estimate benefits between $0.38 billion and $1.41 billion. These results suggest that PMV was cost-effective even under strict assumptions, underscoring the potential for governance-oriented conservation programs to generate scalable and economically meaningful climate benefits.

Our study contributes to three strands of literature. First, we contribute to a growing literature which studies the effects of environmental and conservation policies. Previous work has evaluated the impact of policies such as payments for ecosystem services (Jack et al. 2008; Jayachandran et al. 2017), indigenous land titling (Baragwanath et al. 2020; Baragwanath et al. 2023), and command-and-control policies (Assunção et al. 2023; Assunção et al. 2015). We add to this literature by studying the effects of a voluntary conservation program focused on improving local governance. In particular, our study adds to prior studies which analyze the effects of the PMV on deforestation (Moz-Christofoletti et al. 2022; Sills et al. 2020). We reanalyze the effects of PMV through new data and methods, focusing on additional outcomes such as secondary forest growth, while also digging deeper into the mechanisms driving the results.

Second, we add to a nascent literature on the role of secondary forests in global carbon sinks (Heinrich et al. 2023; Heinrich et al. 2021; Chazdon et al. 2016a). While much work in economics has focused on the effects of conservation policies on primary forest loss, this work overlooks the dynamics of growth and disturbance of secondary forests, which have been shown to hold significant carbon sink potential, with the ability to offset about 26% of emissions from primary forest loss dynamics (Heinrich et al. 2023). We add to a small but growing body of work which seeks to understand when and where secondary forests are allowed to regrow (Baragwanath et al. 2023; Camara et al. 2023; Heinrich et al. 2023), and quantify the contribution of secondary forests to climate change mitigation (Heinrich et al.

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2021; Heinrich et al. 2023; Chazdon 2008).

Following the simple forest transitions model (Mather 1990; Mather 1992; Mather et al. 1998) and the role of governance in the process of forest transitions (Barbier et al. 2017; Barbier et al. 2015), we propose that local institutional factors such as the quality of local governance and investments in monitoring and control play an important role in competing land use decisions (Amacher et al. 2009; Burgess et al. 2019), which can affect not only the rate of forest clearing but also decisions to allow forests to regenerate. Our estimates suggest that the gains in secondary forest growth, when valued using standard carbon prices, may be sufficient to offset program costs, even without considering effects on avoided deforestation. While policies that focus on avoided loss of primary forests should remain a priority for policymakers, designing programs which also protect secondary forests and allow for regeneration on previously disturbed forests can be an important part of the policy agenda.

We contribute to the literature examining decentralization and local governance in environmental management. Theory suggests that decentralization can improve environmental outcomes by bringing decision-making closer to stakeholders (Agrawal et al. 1999; Ostrom 1990), but empirical evidence has been mixed. Some studies find decentralization reduced deforestation (Andersson et al. 2007) and enabled more context-specific environmental policies (Miller et al. 2015), while others show it can exacerbate environmental problems through elite capture or insufficient local capacity (Bardhan et al. 2000; Burgess et al. 2012; Lemos et al. 2007). This inconsistency in outcomes highlights a key insight from political economy: institutional design matters critically for decentralization's effectiveness (Besley et al. 2003; Seabright 1996). Our findings demonstrate that when decentralization includes local capacity building and dedicated funding, as in the PMV case, it can substantially improve environmental management.

The rest of the paper is organized as follows. Section 2 provides some background on the PMV and the institutional environment in which it was enacted. Section 3 presents our methods and data, while Section 4 shows the results from our estimation. Section 6 presents

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our estimated carbon capture effects of PMV and the corresponding estimates of economic benefits from the policy, while Section 7 concludes.

2 Background

Between 2004 and 2014, Brazil emerged as a global leader in combating deforestation, achieving a remarkable 80% reduction in forest clearing rates (Nepstad et al. 2014; Piketty et al. 2015; Antonaccio et al. 2018; Alves-Pinto et al. 2018; Heilmayr et al. 2020). This dramatic decline demonstrates the effectiveness of Brazil's innovative policy approach, which combined command-and-control regulations with voluntary initiatives. The comprehensive strategy included imposing credit restrictions on high-deforestation areas (Souza et al. 2020; Assunção et al. 2020), promoting sustainable supply-chain agreements (Lambin et al. 2018), enhancing satellite monitoring capabilities and strengthening local governance (Assunção et al. 2023; Assunção et al. 2019; Tacconi et al. 2019), and implementing targeted enforcement actions against illegal deforestation (Antonaccio et al. 2018; Assunção et al. 2017).In this paper, we focus on the benefits created by the PMV in particular, which we describe in detail below.

2.1 The Green Municipalities Program - Programa Municípios Verdes (PMV)

The PMV was initiated by the state government of Pará as a voluntary program for municipalities. It was developed by the state government in collaboration with municipalities, civil society, private firms, the Brazilian Institute for the Environment and Renewable Natural Resources (IBAMA), and the Federal Public Ministry (MPF) (P.M.V. 2021a; P.M.V. 2021b). The program's primary objectives were to combat deforestation in the state, boost sustainable rural production, implement strategic environmental and land management actions, and promote local agreements between stakeholders. Secondary objectives included the implementation of the Rural Environmental Registry (CAR), and structuring environmental

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management through the Environmental Compliance Program (Programa de Regularização Ambiental – PRA) in participating municipalities (P.M.V. 2021a; P.M.V. 2021b).

Municipalities join the PMV either by signing a Term of Commitment (TAC) with the Federal Public Ministry or by directly signing a Term of Adhesion (TA) with the state government. These agreements establish a stable legal and political framework for the commitments. Participating municipalities receive various incentives, including environmental clearance licenses, tax benefits, training for local officials, and support for implementing the CAR and subsequent PRA. These incentives aim not only to encourage participation but also to support the broader goal of establishing sustainable financial frameworks for local environmental and resource management (P.M.V. 2021a; P.M.V. 2021b). Figure 1 illustrates the staggered uptake by municipalities between 2010 and 2019, showing that by 2019, 129 of Pará's 144 municipalities were participating (out of 772 municipalities in the Brazilian Legal Amazon according to IBGE).

The PMV program provides a detailed outline consisting of seven crucial steps for municipalities that participate (P.M.V. 2021a; P.M.V. 2021b; Moz-Christofoletti et al. 2022):

1. Initiate a non-binding social agreement with the aim of reducing deforestation rates.
2. Establish a dedicated working group responsible for monitoring deforestation.
3. Register 80% of the land in the municipality, excluding protected areas, in the CAR. This is an essential requirement for municipalities to be removed from the deforestation blacklist.
4. Reduce and maintain deforestation rates below 40 square kilometers annually. This is another prerequisite for being taken off the blacklist.
5. Conduct field checks in response to deforestation alerts.
6. Strive to not be listed on the deforestation blacklist.

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7. Establish an environmental governance structure, which includes a municipal council, a municipal environmental fund, and a system for environmental licensing.

These steps show a clear emphasis on curtailing deforestation, specifically to keep deforestation levels low enough to keep municipalities out of the blacklist. However, in practice PMV extends beyond command-and-control by fostering improvements in local governance and land regularization. Between 2010 and 2018, participating municipalities received: (i) technological support through high-resolution data to issue municipal bulletins for Field Verification of Deforestation (Verificação do Desmatamento em Campo - VDC), (ii) equipment including 4x4 vehicles for field operations, computers, and cameras to support ground-level monitoring, (iii) human capital development with 99 technicians trained in VDC, 1,049 in environmental management, 167 in rural activity licensing, and 50 in environmental management systems, and (iv) land titling support with direct assistance to 4,824 rural properties for CAR registration, resulting in approximately 90,000 hectares being registered (P.M.V. 2021b).

Although previous studies have shown minimal or null effects of PMV on reducing deforestation beyond federal policies (particularly blacklisting) (Sills et al. 2020; Moz-Christofoletti et al. 2022), we revisit this relationship, and hypothesize that PMV participation will yield broader benefits beyond deforestation control, including increased secondary forest growth, reduced land-related conflicts, and higher non-agricultural GDP, stemming from improved environmental monitoring and enhanced land management practices within municipal bureaucracies. However, acknowledging that blacklisting plays a major role in the policy environment, we investigate to what extent blacklisting is driving our results by eliminating from our sample of treatment and control municipalities any municipality that was ever blacklisted.

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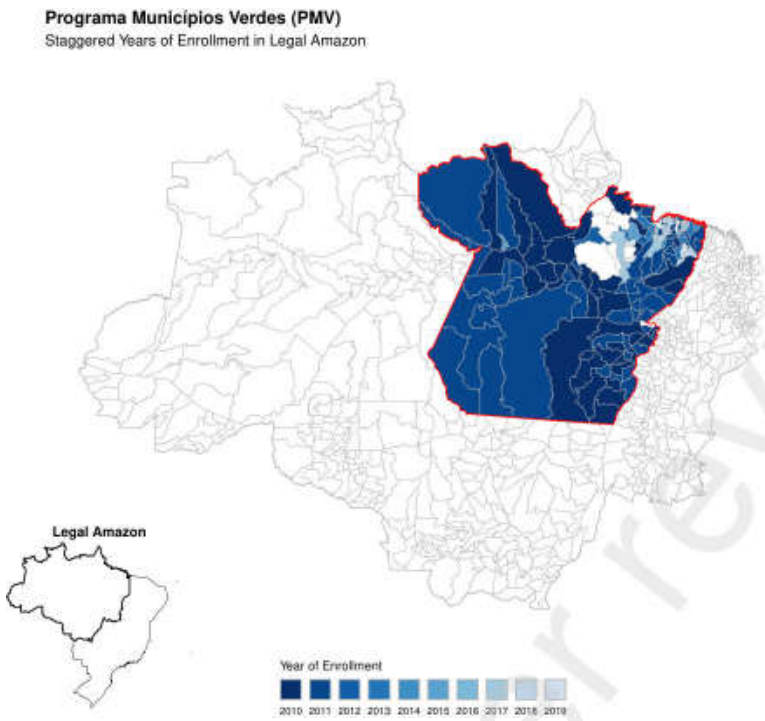


Figure 1: Timing of municipal enrollment in the PMV Program from 2010 to 2019. Staggered participation of municipalities in is shown in the legend where lighter shades represent earlier adopters. The state of Pará is highlighted with a red border.

2.2 A Note on the Blacklisting of Brazilian Municipalities (Municípios Prioritários)

The Priority Municipalities program (Municípios Prioritários–MPs), commonly known as the Blacklisting program, was implemented by the Brazilian government in 2008 to combat Amazon deforestation (Arima et al. 2014; Assunção et al. 2019). Established under the federal Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm), the program aimed to reduce deforestation by 80% by 2020 compared to the 1996-2005 baseline (West et al. 2021).

Municipalities with high deforestation rates and environmental violations were designated as "priority municipalities" based on satellite monitoring data from Brazil's National Institute for Space Research (INPE). Blacklisted municipalities faced three primary consequences: (i) intensified enforcement of environmental regulations by IBAMA, resulting in increased

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fines for illegal deforestation (Börner et al. 2015); (ii) restricted access to agricultural credit for non-compliant landowners (Arima et al. 2014); and (iii) suspension of new rural property registrations, limiting landowners' access to legal documentation and resources (Assunção et al. 2019). Figure C.3 shows a map of the staggered entry into the blacklist.

Multiple studies have documented significant deforestation reductions in blacklisted municipalities compared to non-blacklisted ones (Arima et al. 2014; Cisneros et al. 2015; Assunção et al. 2019). Our paper does not aim to test the Blacklisting program's effects but recognizes it as the critical policy backdrop for our study of the Green Municipality Program (PMV). The threat of blacklisting remained a constant concern for Legal Amazon municipalities, making PMV's effects inseparable from this context. As shown previously, PMV was designed around blacklisting thresholds, with many target outcomes directly linked to criteria that would trigger blacklisting. Therefore, we analyze PMV's effects as supplementary to these existing federal command-and-control policies

3 Data and Methods

3.1 Data

To test the effects of PMV, we rely on both satellite derived data and administrative data at the municipality level within the Brazilian Amazon. Our analysis focuses on three main dependent variables:

Deforestation (PRODES): We use the annual deforestation measurements from the Brazilian National Institute for Space Research's PRODES system, which provides high-resolution satellite monitoring of forest cover loss in the Brazilian Amazon. These measurements are considered the official reference for deforestation tracking in Brazil and are reported in square kilometers per year ($km^2/year$) at the municipality level.

Deforestation (Mapbiomas): As a complementary measure of deforestation, we incorporate data from the Mapbiomas project, which uses Landsat satellite imagery to track land

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use and land cover changes. Unlike PRODES, Mapbiomas captures smaller-scale forest disturbances and provides annual deforestation estimates in square kilometers ($km^2/year$) at the municipality level.

Secondary Forest Growth: To measure secondary forest growth, we use novel data by Junior et al. 2020 which relies on Landsat satellite imagery to identify and track the extent, age, and dynamics of secondary forest regrowth across the Brazilian Amazon from 1985 to 2017. This dataset provides unprecedented spatial and temporal resolution for monitoring forest regeneration patterns following deforestation events. Our panel includes total secondary forest area ($km^2/year$) and area of secondary forest by age group ($km^2/year$) at the municipality level, categorized into three age classes: young (1-5 years), intermediate (6-15 years), and mature (16-34 years).

Covariates: Our covariates include climatic, economic, and policy enforcement-related variables. Climatic factors are considered primary drivers of forest loss and recovery (Zeng et al. 2019). Previous research has highlighted the local impact of precipitation and temperature variation on deforestation (Anderson et al. 2018), and these factors are widely recognized as primary drivers affecting secondary forest recovery (Zeng et al. 2019). We include municipality-level yearly mean annual precipitation (mm) and temperature (K) constructed using Google Earth Engine (GEE) from the TerraClimate database (Abatzoglou et al. 2018).To effectively control for local economic activity, we include measures of municipal gross domestic product (GDP). This measure reflects the total value of all goods and services produced within the local economy. We obtain GDP measures by aggregating the gross value added (GVA) of all agriculture, industries, and services operating within a municipality from IBGE[3]. Furthermore, population density has been found to be a critical factor for both economic outcomes and forest outcomes (Mullan et al. 2018; Gori Maia et al. 2020). We measure population density by dividing the number of inhabitants by the area of the municipality. Lastly, we include the area of protected land (under Protected Areas,

[3]For more detailed information on these estimates, please refer to (Instituto Brasileiro de Geografia e Estatística (IBGE) 2021).

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National Parks, or Indigenous Territories) within a given municipality, which plays a crucial role in assessing forest protection policy enforcement. This variable indicates both the availability of land for protection and the level of monitoring within the municipality, as multiple agencies may be involved in forest protection.

Our treated group of municipalities is defined based on a dummy variable that represents whether the municipality chooses to join the PMV program at any point in time (i.e., it takes the value of 1 if the municipality ever decided to join the Program and 0 otherwise). From 2010 to 2019, the PMV program gradually attracted 90% (129 out of 144) of the municipalities in the state of Par'a. The untreated group comprises the 629 non-participating municipalities located throughout the Brazilian Legal Amazon.

Table 1 shows summary statistics for our data by treatment (PMV participating municipalities) and control groups. Upon inspection, it becomes clear that participating municipalities are significantly different to non-participating municipalities. From Table 1, we see that PMV municipalities have significantly higher levels of deforestation (by both PRODES and Mapbiomas measurements), significantly higher levels of secondary forests across all age categories, higher rates of environmental fines, as well as larger amounts of area under protected area status. Interestingly, they have a significantly lower area registered in the CAR, which is one of the objectives included in the PMV. These systematic differences between treatment and control provide motivation for the use of the generalized synthetic controls method, which we describe in detail below.

3.2 Method: Generalized Synthetic Controls

To identify the causal impact of PMV on deforestation and secondary forest recovery, we employ the Generalized Synthetic Control method with Expectation-Maximization estimation (GSC-EM) (Xu 2017b). This method addresses key identification challenges in our setting: the voluntary nature of PMV enrollment, staggered adoption across municipalities, and the need to construct credible counterfactuals when treatment assignment is non-random. The

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Table 1: Summary Statistics

		Non PMV		PMV	
Variable Name	Obs.	Mean (SD)	Obs.	Mean (SD)	Test
Deforestation PRODES (Km^2)	11951	14.33 (55.42)	2451	39.45 (110.56)	F=270.7***
Deforestation Mapbiomas (Km^2)	11951	26.39 (52.90)	2451	53.37 (104.04)	F=128.4***
Secondary Forest (Km^2)	11951	153.37 (208.81)	2451	399.49 (418.21)	F=1868.5***
Secondary Forest Age 1-5 (Km^2)	11951	68.14 (103.52)	2451	176.45 (195.52)	F=1549.6***
Secondary Forest Age 6-15 (Km^2)	11951	54.50 (77.83)	2451	142.75 (154.69)	F=1741.3***
Secondary Forest Age 16-34 (Km^2)	11951	31.14 (59.47)	2451	81.09 (118.82)	F=950.7***
Environmental Fines (2019R$ '000)	11951	4086.0 (28090.70)	2451	13767.39 (77024.34)	F=114.5***
Non-agriculture GVA (2019R$ '000,000)	10693	298.0 (195.55)	2193	526.10 (2024.32)	F=24.5***
Precipitation (mm)	11951	151.68 (43.68)	2451	196.81 (46.59)	F=2121.4***
Temperature (K)	11951	317.76 (9.37)	2451	315.76 (6.30)	F=102.4***
Protected Area (Km^2)	11951	144247.25 (572381.91)	2451	215784.30 (898660.11)	F=25.4***
CAR Area (Km^2)	11951	11740.72 (95103.36)	2451	279.19 (5597.94)	F=25.4***

Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$
NOTE: All secondary forest area are yearly. Gross-value added by non-agriculture activity at municipality level at current prices, Environmental Fines are sum of all fines by IBAMA in given municipality by year,

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PMV program's implementation presents substantial identification challenges. First, participation is voluntary, creating selection concerns as municipalities that choose to enroll may differ systematically from non-participants in both observable and unobservable characteristics. Second, municipalities joined the program at different times, requiring methods that can accommodate staggered treatment adoption. Third, the program was implemented exclusively in Pará state, potentially limiting the pool of suitable control units from other Amazonian regions.

GSC-EM is particularly well-suited to address these challenges by relaxing the parallel trends assumption required in conventional difference-in-differences approaches. The method explicitly models time-varying unobserved confounders through a factor structure that captures heterogeneous trends across municipalities. This allows us to construct more credible counterfactuals for treated municipalities by accounting for their pre-treatment trajectories and structural differences from non-treated units.

The functional form of our estimation using the GSC-EM method is:

$$Y_{mt} = \beta_{mt} D_{mt} + X'_{mt}\theta + \lambda'_m f_t + \epsilon_{mt} \tag{1}$$

where Y_{mt} represents our outcome variables (deforestation or secondary forest area) for municipality m in year t; D_{mt} equals 1 if municipality m in year t has enrolled in the PMV program and 0 otherwise; and β_{mt} represents the heterogeneous treatment effect. The vector X_{mt} includes observable control covariates such as precipitation, temperature, GDP per capita, population density, and protected area coverage, with θ representing their effects.

The term $\lambda'_m f_t$ captures unobserved heterogeneity, where f_t is a vector of unobserved common factors and λ_m is a vector of municipality-specific factor loadings. This component accommodates various forms of unobserved heterogeneity, including municipality and year fixed effects, municipality-specific time trends, and other confounding factors that can be decomposed into this interactive structure. Finally, ϵ_{mt} represents idiosyncratic disturbances

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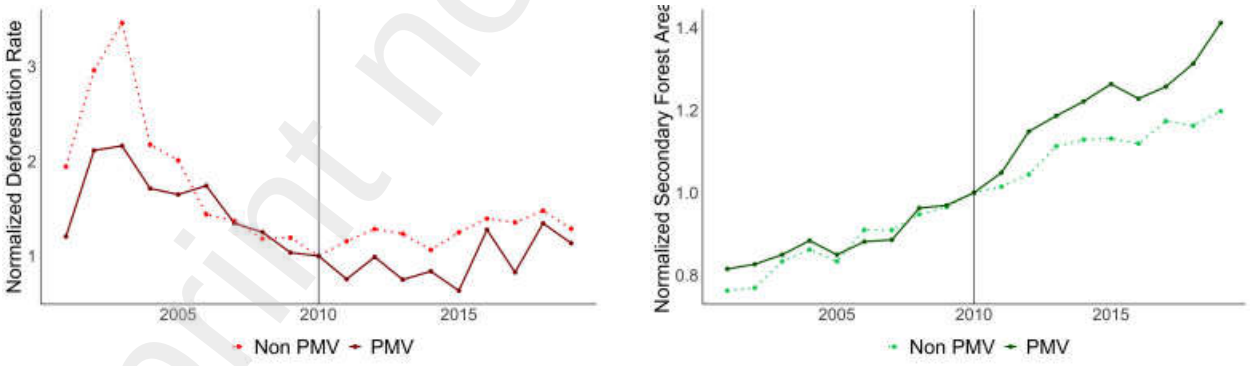
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with zero mean.

The average treatment effect on the treated (ATT) is calculated by averaging β_{mt} across all treated municipalities and post-treatment periods. We implement this estimation using cross-validation to select the optimal number of latent factors, reducing the risk of overfitting while maintaining model flexibility. In supplementary analyses (in Appendix G.1.1), we confirm the robustness of our findings using alternative specifications, including two-way fixed effects models, the Callaway and Sant'Anna (Callaway et al. 2021) staggered difference-in-differences estimator, and interactive fixed effects approaches (Xu 2022). These additional analyses consistently support our main findings, though the GSC-EM estimates tend to be more conservative, yielding the lowest treatment effects among all methods considered—further reinforcing their reliability.

4 Results

4.1 Preliminary Evidence on the Effects of PMV on Deforestation and Secondary Forest Growth



(a) Deforestation (b) Secondary Forests

Figure 2: Trends in Deforestation and Secondary Forest by PMV participation

The preliminary evidence presented in Figure 2 illustrates the contrasting trends in deforestation rates and secondary forest growth between municipalities participating in the Green

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Municipality Program (PMV) and non-participating municipalities. The figures provide a visual comparison of normalized deforestation rates (Figure 2, Panel (a)) and normalized secondary forest growth (Figure 2, Panel (b)) before and after the PMV implementation in 2010.

As demonstrated in Figure 2, Panel (a), normalized deforestation rates between PMV and non-PMV municipalities followed similar pre-trends before the policy's introduction, indicating parallel trajectories. Both groups experienced a gradual decline in deforestation leading up to 2010. However, post-2010, the trends diverge significantly. Deforestation rates in PMV municipalities continue to decrease more substantially compared to non-PMV municipalities, which exhibit relatively stable or slightly increasing deforestation rates. This divergence suggests the PMV has effectively reduced deforestation beyond the general declining trend observed across the region.

Figure 2, Panel (b) depicts the trends in secondary forest growth for both municipality types. Similar to deforestation rates, the pre-trends for secondary forest growth are comparable, indicating no significant differences between the two groups before PMV implementation. After 2010, however, a notable divergence emerges. PMV municipalities experience substantial increases in secondary forest growth, while growth in non-PMV municipalities remains comparatively modest. This significant post-2010 distinction highlights the PMV's impact on promoting forest recovery.

The pronounced difference in post-2010 trajectories for both deforestation rates and secondary forest growth offers preliminary evidence suggestive of PMV's potential effectiveness in achieving its conservation objectives. The more substantial impact on secondary forest growth underscores the importance of considering this outcome when evaluating the environmental benefits of the policy.

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4.2 Effect of PMV on Deforestation and Secondary Forests

Table 2 presents the impact of the Green Municipality Program (PMV) on deforestation and secondary forest area, using the generalized synthetic control (GSC-EM) method. Columns (1) and (2) focus on deforestation outcomes from the PRODES and MapBiomas datasets, respectively, while columns (3) through (6) examine secondary forest area, disaggregated by forest age.

In the full sample (Panel A), participation in PMV is associated with a 44% reduction in deforestation based on PRODES data (Column 1) and a 12% reduction using MapBiomas data (Column 2). These estimates are statistically significant, though the difference in magnitude reflects underlying differences between the datasets: PRODES focuses on large-scale clear-cutting of primary forest, while MapBiomas captures smaller-scale and fragmented changes in land cover. Throughout this analysis, we place more weight on results measured by MapBiomas due to its higher spatial resolution and greater sensitivity to small-area disturbances, which are more likely to respond to local governance interventions.[4]

However, caution is warranted in interpreting these patterns as the causal effect of PMV alone. Many treated municipalities were simultaneously subject to federal blacklisting, a command-and-control policy that may have influenced deforestation independently. When we exclude all blacklisted municipalities (Table 2, Panel B), the PRODES-based estimates remain negative and significant, but the effect using MapBiomas data becomes smaller and statistically insignificant. This sensitivity underscores the difficulty of disentangling the effects of PMV from broader federal enforcement dynamics, particularly when it comes to deforestation outcomes.

Event-study plots in Figure 3 illustrate the dynamics of deforestation before and after PMV implementation, separately displaying estimates for both the full sample (blue coefficients) and the restricted sample, which excludes municipalities that were ever blacklisted by federal enforcement (orange coefficients). Panel (a), using PRODES data, shows a pro-

[4]For a detailed discussion of differences between PRODES and MapBiomas, see Baragwanath et al. 2025.

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gressive decline in deforestation relative to synthetic controls, with increasingly negative post-treatment estimates and no evidence of differential pre-trends, supporting the validity of the counterfactual scenario. Panel (b), utilizing MapBiomas data, similarly reveals an immediate reduction in deforestation following PMV implementation, though the magnitude of this effect diminishes approximately five years post-treatment. Estimates for the restricted sample are consistently slightly smaller, yet they exhibit a similar overall trend, highlighting the robustness of these findings even when controlling for concurrent federal enforcement interventions

In contrast, the association between PMV and secondary forest growth is more robust across specifications. In the full sample, we estimate a 7% increase in total secondary forest area (Column 3), with most of this effect concentrated in young forests (1–8 years old), which increase by 11% (Column 4). Effects on middle-aged and older forests are smaller in magnitude—1.2% and 3.0%, respectively (Columns 5–6)—consistent with the longer time required for forest maturation. Event-study plots in Figure 4 show no evidence of differential pre-trends and suggest that most of the increases in forest cover occur after PMV implementation, particularly for younger forests.

The positive association between PMV and secondary forest growth persists even when blacklisted municipalities are excluded (Panel B). In this restricted sample, we estimate a 5% increase in total secondary forest cover, again concentrated in young forests, with smaller gains in middle-aged and older forests. These results are consistent with the interpretation that PMV contributed to forest recovery by altering land use incentives and strengthening local governance, particularly in municipalities where federal enforcement pressure was lower.

Overall, our findings indicate that PMV participation is associated with reductions in deforestation—though these results are more dependent on concurrent federal interventions—and more consistently associated with increases in secondary forest cover. The latter effects are robust to sample restrictions and persist across forest age classes, suggesting that local governance programs may play a meaningful role in facilitating forest regeneration,

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even in the absence of direct enforcement pressure.

Table 2: Effects of PMV on Deforestation and Secondary Forests

Panel A: Full Sample Estimates Using Generalized Synthetic Control (GSC-EM)						
	Deforestation Area		**Secondary Forest Area**			
	PRODES	MapBiomas	Total	Young Forest	Middle Forest	Old Forest
	(1)	(2)	(3)	(4)	(5)	(6)
ATT	-0.44***	-0.12***	0.07***	0.11***	0.01*	0.03*
	(0.03)	(0.04)	(0.00)	(0.01)	(0.00)	(0.01)
Treated Munics	124	124	124	124	124	124
Control Munics	634	634	634	634	634	634
Periods	17	17	17	17	17	17
Panel B: Restricted Sample Estimates Excluding All Municipalities Ever Blacklisted (GSC-EM)						
	Deforestation Area		**Secondary Forest Area**			
	PRODES	MapBiomas	Total	Young Forest	Middle Forest	Old Forest
	(7)	(8)	(9)	(10)	(11)	(12)
ATT	-0.47***	-0.07	0.05***	0.06***	0.01***	0.05***
	(0.03)	(0.05)	(0.00)	(0.01)	(0.00)	(0.01)
Treated Munics	111	111	111	111	111	111
Control Munics	604	604	604	604	604	604
Periods	17	17	17	17	17	17

Notes: Panel A presents results from the Generalized Synthetic Control (GSC-EM) method using the full sample. Columns (1) and (2) report effects on the log rate of deforestation using PRODES and MapBiomas data, respectively. Columns (3) to (6) show the effect on secondary forest area, including decomposition by forest age: Young (1–8 yrs), Middle-aged (9–15 yrs), and Old (16+ yrs). Panel B replicates this analysis using only municipalities that were never blacklisted by IBAMA, showing only the GSC-EM estimates. All regressions include municipality and year fixed effects. Covariates include mean precipitation, mean temperature, GDP per capita, population density, and proportion of land covered by protected areas. Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

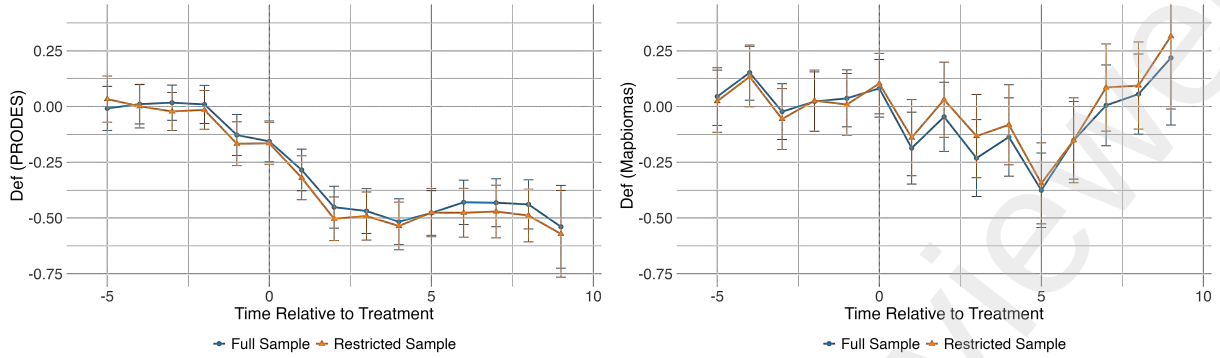
5 Mechanisms: Land Regularization, Enforcement, Land Conflict, and Economic Effects

This section explores potential causal pathways through which the Green Municipality Program (PMV) may have influenced environmental and socioeconomic outcomes in participat-

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Figure 3: Effects of PMV on Deforestation



(a) PRODES (b) MapBiomas

Notes: Figures show results plots from different methods. The dependent variable is the log of deforestation, and the model includes municipality and year fixed effects. The results are using the Generalized Synthetic Controls methods via EM model using covariates. Panel (a) shows results for PRODES dataset and Panel (b) shows results using MapBiomas dataset. Covariates included are mean precipitation, mean temperature, GDP per capita, population density, and proportion of land covered by Protected Areas.

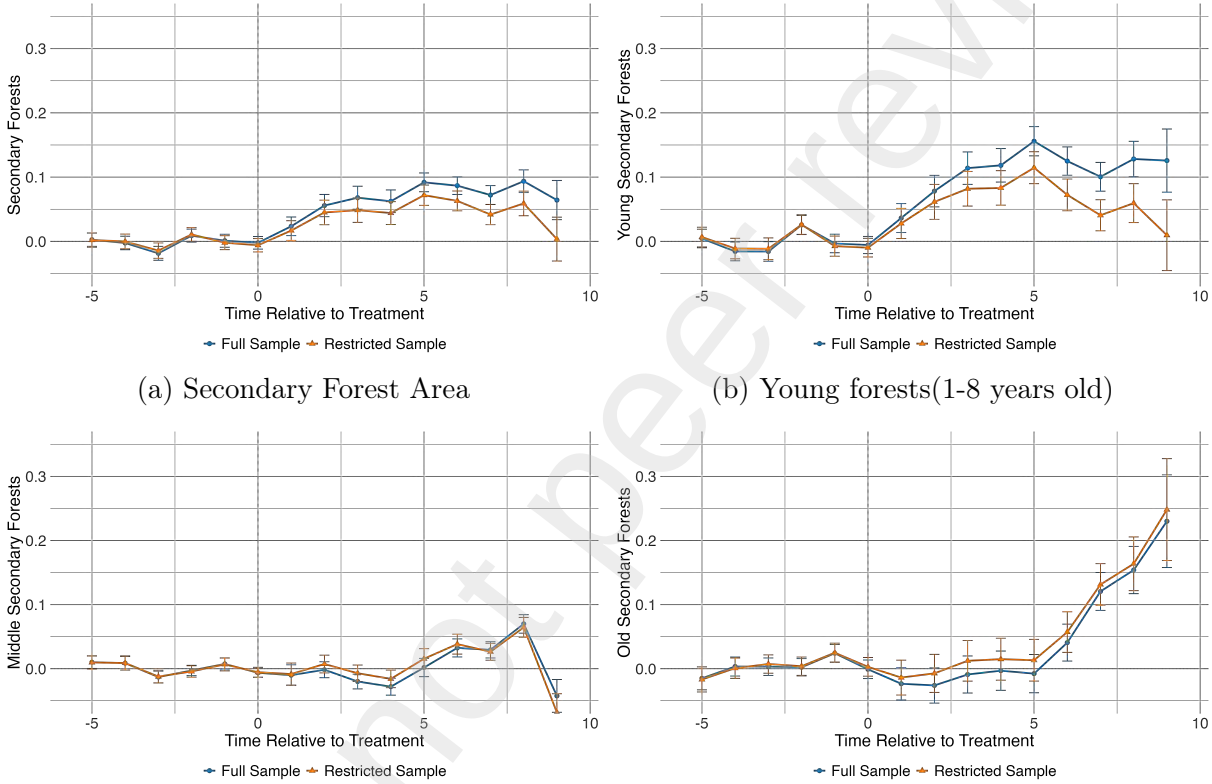
ing municipalities. We propose several plausible mechanisms that could explain the observed effects on deforestation reduction and secondary forest expansion. First, we hypothesize that land regularization through the Rural Environmental Registry (CAR) played a crucial role in altering landholders' incentives, particularly for medium and large-scale properties. By clarifying land tenure and increasing monitoring capacity, PMV may have encouraged landholders to reduce deforestation and allow previously cleared land to regenerate, facilitating secondary forest recovery.

Second, we posit that PMV's investments in local governance capacity—including equipment provision, personnel training, and institutional development—enhanced enforcement effectiveness, potentially reducing illegal land clearing activities and associated phenomena such as forest fires and land conflicts. Third, we conjecture that PMV's emphasis on sustainable rural production may have facilitated economic diversification away from deforestation-intensive activities, potentially increasing non-agricultural economic output. Finally, we suggest that PMV's requirements for establishing environmental governance structures likely strengthened municipal environmental management capacity through increased staffing and dedicated institutional arrangements.

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Figure 4: Effects of PMV on Secondary Forests



(a) Secondary Forest Area

(b) Young forests(1-8 years old)

(c) Middle-aged forests (9-15 years old)

(d) Old forests (16+ years old)

Notes: Figures show results plots from different models. The dependent variable is the log of secondary forest extent, and the model includes municipality and year fixed effects. Panel (a) shows results for a overall secondary forest area, Panel (b) shows results for Young forests (1-8 years old), Panel (c) shows results for Middle-aged forests (9-15 years old) and Panel (d) shows results for and Old forests (16+ years old). Covariates included are mean precipitation, mean temperature, GDP per capita, population density, and proportion of land covered by Protected Areas.

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The following subsections evaluate these hypotheses empirically, examining: (1) whether land regularization through CAR registration explains the program's effects on forest outcomes; (2) how enforcement, conflict, and economic activity changed in response to PMV implementation; and (3) whether municipal administrative capacity expanded in treated municipalities.

5.1 Mechanisms I: Land Regularization

This section examines the hypothesis that PMV's effects on secondary forest expansion are mediated through land regularization, specifically via the Rural Environmental Registry (CAR). If improved land regularization drives forest outcomes, we would expect to observe more pronounced effects within formally registered properties compared to undesignated lands.

The CAR, established under Brazil's 2012 Forest Code, functions as a national registry for rural properties that serves as a critical instrument for deforestation monitoring and environmental compliance enforcement. Among PMV's core objectives, CAR implementation featured prominently in the requirements for municipalities to attain and maintain "Green Municipality" status. During program implementation, PMV provided substantial technical support for CAR registration, assisting 4,824 rural properties that ultimately registered approximately 90,000 hectares into the system (P.M.V. 2021a). Qualitative research, including interviews with municipal administrators, indicates that technical assistance for georeferencing and CAR registration represented one of the most tangible manifestations of PMV within municipal bureaucracies (Da Costa *et al* 2015).

To isolate and analyze forest dynamics specifically within CAR-registered properties, we use shapefiles containing boundaries of all properties registered in the CAR system as of 2022. This dataset encompasses over 800,000 rural properties across the Legal Amazon, covering approximately 3,428,055.5 square kilometers (representing 66% of the Legal Amazon's total area). By intersecting these property boundaries with MapBiomas forest cover data, we

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calculate deforestation and secondary forest growth at the property level, then aggregate these measurements to the municipality level for analytical purposes.

We acknowledge an important limitation in our approach: the CAR property boundaries represent a cross-sectional snapshot as of 2022, without historical registration dates. Consequently, when analyzing forest dynamics across our study period, we cannot definitively determine whether specific land parcels maintained consistent ownership or were even formally designated as rural properties throughout the entire timeframe. Despite this limitation, restricting our analysis to areas within CAR boundaries provides valuable insights into whether PMV's effects predominantly manifest on lands formally claimed and managed as private property, as opposed to undesignated public lands, protected areas, indigenous territories, or other categories of protected public lands.

To test this mechanism, we run our regressions on forest outcomes within CAR-registered properties at the municipality level. If CAR implementation in fact strengthens environmental management and facilitates more effective law enforcement, we would expect the majority of PMV's effects to be driven by land within these formally registered properties rather than dispersed across all land types.

Table 3, Panel A presents the effects of PMV on deforestation and secondary forest area within CAR properties. The first row reports the impact on deforestation, where the dependent variable is the mean percentage of forest loss within CAR-registered properties as measured by Mapbiomas. The second row presents the impact on secondary forests, where the dependent variable is the percentage increase in secondary forest area within CAR properties. For both deforestation and secondary forests, we first estimate the effects across all CAR-registered properties and then separately for small ($\leq$ 20 ha), medium (20–125 ha), and large (>125 ha) properties. This breakdown allows us to identify whether the effects of PMV vary by property size, shedding light on whether land tenure security and environmental enforcement had differential impacts across smallholders and larger landowners.

From Panel A, we observe that PMV led to a 3% reduction in deforestation within

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Table 3: Effects of PMV within CAR Properties and Mechanisms

	(1) All	(2) Small	(3) Medium	(4) Large
Panel A: Effects within CAR Properties by Property Size				
ATT Deforestation CAR	-0.03***	0.00	-0.02***	-0.01
	(0.00)	(0.00)	(0.00)	(0.01)
ATT Secondary Forest CAR	0.40***	0.04	0.23***	0.07***
	(0.07)	(0.04)	(0.03)	(0.01)
Treated Munics	123	123	123	123
Control Munics	618	618	618	618
Periods	17	17	17	17
Panel B: Mechanisms - Effects on Enforcement, Conflict, and Economic Activity				
	Fines	Fires	Land Conflict	Non-Ag Value
ATT	-0.69***	-31.98***	-0.02***	0.05***
	(0.17)	(2.96)	(0.00)	(0.00)
Treated Munics	123	123	123	123
Control Munics	618	618	618	618
Periods	17	17	17	17

Notes: Table presents results from the Generalized Synthetic Control Method using the EM approach (Xu 2017a). Panel A examines the impact of PMV on deforestation and secondary forest growth within CAR-registered properties of different sizes: small ($\leq$ 20 ha), medium (20–125 ha), and large (>125 ha). Panel B investigates potential mechanisms through which PMV influences deforestation and secondary forest expansion, including environmental fines, fire occurrences, land conflicts, and non-agricultural economic growth. All models control for municipality and year fixed effects, as well as covariates such as mean precipitation, mean temperature, GDP per capita, population density, and the proportion of land covered by Protected Areas.
Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

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CAR properties, confirming that land tenure regularization and improved environmental monitoring under CAR likely contributed to preventing illegal deforestation. The effects are not uniform across all property sizes, with the largest impact seen in medium-sized properties, which experienced a 2% decrease in deforestation. This suggests that medium landholders were particularly responsive to CAR enforcement and tenure security, possibly because they are large enough to be subject to monitoring but not as influential as large landholders who might have more resources to resist regulation.

More importantly, PMV had a substantial effect on secondary forest expansion within CAR properties, indicating that landholders in treated municipalities not only reduced deforestation but also allowed previously cleared land to regenerate. Across all CAR-registered properties, secondary forest cover increased by 40%, a significant shift towards reforestation. Medium-sized properties accounted for the bulk of this effect, with secondary forest cover expanding by 23%, depending on the specification. Large properties also saw an increase, though at a lower magnitude, with secondary forests expanding by approximately 7% within their boundaries.

The findings from Panel A provide strong evidence that land regularization through CAR played a crucial role in reducing deforestation and fostering secondary forest growth within PMV municipalities. The results suggest that medium-sized properties were the most responsive to the program, possibly due to stronger enforcement mechanisms, greater incentives to comply with environmental regulations, and better access to technical assistance for reforestation.

5.2 Mechanisms II: Enforcement, Land Conflict, and Economic Effects

We next examine how PMV's investments in local governance capacity might explain the observed secondary forest expansion through four interrelated mechanisms: environmental enforcement, fire reduction, conflict mitigation, and economic diversification.

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If PMV improved monitoring capacity through better equipment and trained personnel, we would expect changes in enforcement outcomes. While fire reduction should follow from enhanced monitoring, the effect on environmental fines is theoretically ambiguous—better detection could increase fines, but awareness of stronger enforcement might deter violations, reducing fines. Given PMV's participatory approach, involving multiple stakeholders, and the accountability mechanisms created through CAR implementation, we hypothesize that the deterrence effect would dominate, leading to fewer environmental violations and thus fewer fines.

Panel B of Table 3 confirms this prediction, showing PMV reduced environmental fines by 69% per municipality annually. This decline suggests heightened compliance with environmental regulations in participating municipalities. As monitoring improved and CAR expanded, deforestation activities became both more detectable and more attributable to specific landowners, reinforcing legal compliance.

PMV also significantly reduced fire incidence, with 31.98% fewer fires annually in participating municipalities. As fires typically precede agricultural expansion in the Amazon, this reduction indicates decreased forest conversion activity. The decline in fire usage reflects both improved monitoring capabilities and changed landholder behavior in response to enforcement risk.

Land conflict, often associated with contested property rights in the Amazon, decreased by 0.02% annually in PMV municipalities. Though modest in magnitude, this reduction is meaningful in a region where land disputes frequently involve violence. By formalizing property boundaries through CAR and providing an institutional framework for resolving claims, PMV appears to have reduced land tenure uncertainty, mitigating a key driver of conflict. Finally, PMV participation increased non-agricultural GDP by 5% per municipality. This suggests economic diversification beyond forest-clearing activities like cattle ranching and logging toward sectors less dependent on deforestation. Improved governance and reduced conflict likely created more favorable conditions for business development in these alternative

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sectors.

To contextualize PMV's effects, we compare them with the outcomes of Brazil's federal blacklisting policy. Appendix Table I.9 presents estimates where we use blacklisting as the treatment. While both policies are associated with reductions in fire incidence, their institutional and economic implications diverge. Unlike PMV, blacklisting increased environmental fines—consistent with a punitive, top-down enforcement approach—and had no discernible impact on land conflict. We also find suggestive evidence that blacklisting may have reduced non-agricultural GDP, potentially due to restricted credit access or heightened uncertainty in affected municipalities. These contrasts underscore PMV's stronger association with governance improvements and local economic diversification, pointing to the potential complementarity between centralized enforcement and decentralized capacity-building.

Taken together, these findings suggest that PMV's influence extended beyond land use outcomes to broader institutional change. By building municipal capacity for environmental management and reducing tenure-related conflict, PMV created a more enabling environment for low-deforestation development. The program's association with improved enforcement, increased tenure security, and a shift toward non-agricultural economic activity aligns with a governance-centered model of conservation. These results highlight how environmental policies that invest in local institutions—rather than relying solely on regulatory pressure—can generate durable co-benefits that support both conservation and development objectives.

6 Quantifying the Economic Benefits of PMV

In this section, we evaluate the biophysical and economic benefits of the Green Municipality Program (PMV) by comparing results from the full sample of municipalities (including those ever blacklisted) with a restricted sample that excludes these municipalities. Specifically, we estimate the impact of PMV on avoided deforestation and secondary forest recovery, and then translate these outcomes into avoided CO_2 emissions and added CO_2 removals. Using

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different social cost of carbon (SCC) values, we convert these carbon outcomes into monetary terms to assess the program's climate-related economic benefits.

Table 4 presents these results. Panel A reports the area-based and carbon-based outcomes for the full sample of municipalities (including blacklisted) over a 10-year post-treatment period. The implementation of PMV led to an estimated 5,683.68 km^2 of avoided deforestation and 36,663.06 km^2 of secondary forest recovery. These land use changes are equivalent to substantial climate benefits: 162.36 million metric tons of avoided CO_2 emissions (i.e., $MtCO_2e$) and an additional 18.35 $MtCO_2e$ removals due to regrowing forests. For perspective, these avoided emissions represent about 7.5% of Brazil's total emissions in 2020.

Panel B converts these carbon outcomes into economic benefits using a range of SCC values from 50 USD to 200 USD per ton of CO_2e. For example, at an SCC of 50 USD, the avoided emissions yield a net present value (NPV) of 6.27 billion USD, and the additional removals generate an NPV of 0.71 billion USD, both discounted at 5% with a 2020 base year. At the high-end SCC of 185 USD—recently recommended by Rennert et al. (2022)—the NPVs rise to 23.19 billion USD from avoided emissions and 2.64 billion USD from added removals. These results suggest that both the emissions avoided and the removals from forest recovery contribute substantially to the economic case for the PMV.

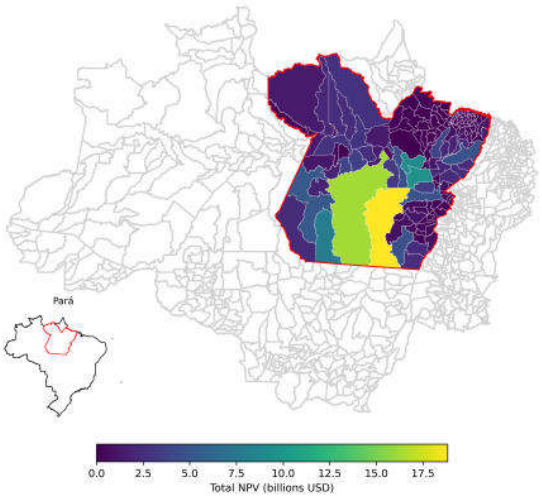
We then assess these benefits using a restricted sample of municipalities that excludes those ever blacklisted, as reported in Panels C and D. This allows us to isolate the effect of the PMV on municipalities that were not subject to potentially confounding federal enforcement mechanisms. In this restricted sample, the program did not yield a statistically significant effect on deforestation, leading to no associated avoided emissions. However, the PMV still produced strong effects on forest recovery: 19,620.39 km^2 of secondary forest regrowth, corresponding to 9.81 million metric tons of CO_2 removals.

Panel D presents the economic benefits from these removals in the restricted sample. At an SCC of 50 USD, the NPV of carbon removals is 0.38 billion USD. This increases to 0.76 billion USD at an SCC of 100 USD, and reaches 1.14 billion USD at an SCC of

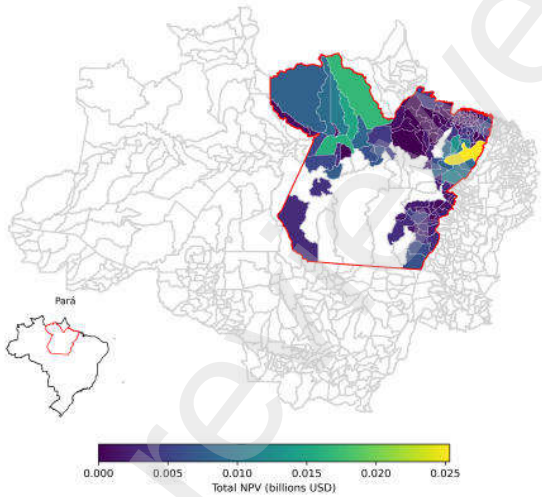
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Figure 5: NPV of PMV from Avoided Emissions and Removals



(a) Full sample (includes ever blacklisted)



(b) Restricted sample (excludes ever blacklisted)

Notes: Maps show the net present value (NPV) of avoided CO_2 emissions and enhanced removals due to PMV implementation, measured in billions of USD. Estimates are computed at the municipality level. Panel (a) uses the full sample, including municipalities ever blacklisted; Panel (b) uses the restricted sample, excluding them.

185 USD. At the upper-bound SCC of 200 USD, the total NPV is 1.52 billion USD. While these numbers are lower than those from the full sample—as expected due to the exclusion of avoided deforestation—they nonetheless highlight the value of carbon sequestration from forest recovery alone.

In summary, the PMV produces substantial environmental and economic benefits, even when focusing only on municipalities not targeted by IBAMA's blacklist. While the restricted sample shows no measurable effect on deforestation—possibly due to policy leakage or selection bias—the forest recovery effects remain robust. These results reinforce the importance of preserving and scaling up support for forest regrowth mechanisms under the PMV framework, especially given their significant carbon removal potential and associated economic returns.

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Table 4: Biophysical and Economic Benefits of PMV: Full vs. Restricted Sample

Panel A: Full Sample – Area and Emissions			
(1) Avoided Deforestation (km^2)	(2) Secondary Forest Recovery (km^2)	(3) Avoided Emissions ($MtCO_2e$)	(4) Added Removals ($MtCO_2e$)
5,683.68	36,663.06	162.36	18.35
Panel B: Full Sample – Economic Benefits (billions USD)			
SCC ($/tCO$_2$e)	(1) NPV Avoided Emissions	(2) NPV Removals	(3) Total NPV
50	6.27	0.71	6.98
100	12.53	1.43	13.96
150	18.80	2.14	20.94
185	23.19	2.64	25.83
200	25.07	2.85	27.92
Panel C: Restricted Sample – Area and Emissions			
(1) Avoided Deforestation (km^2)	(2) Secondary Forest Recovery (km^2)	(3) Avoided Emissions ($MtCO_2e$)	(4) Added Removals ($MtCO_2e$)
0.00	19,620.39	0.00	9.81
Panel D: Restricted Sample – Economic Benefits (billions USD)			
SCC ($/tCO$_2$e)	(1) NPV Avoided Emissions	(2) NPV Removals	(3) Total NPV
50	0.00	0.38	0.38
100	0.00	0.76	0.76
150	0.00	1.14	1.14
185	0.00	1.41	1.41
200	0.00	1.52	1.52

Notes: This table compares the biophysical and economic benefits of PMV implementation between the full sample of municipalities and a restricted sample that excludes ever-blacklisted municipalities. Panels A and C report avoided deforestation, forest recovery (in km^2), and corresponding carbon outcomes (in MtCO$_2$e). Panels B and D present Net Present Values (NPV) under different Social Cost of Carbon (SCC) assumptions. All monetary values are in 2020 USD; NPV is discounted at 5% with base year 2020.

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7 Conclusion

In this paper, we analyze the effects of the Green Municipalities Program (PMV) within the context of the Brazilian Amazon. We find that PMV participation was associated with reduced deforestation and increased secondary forest growth. However, deforestation effects appear to be highly dependent on overlap with federal blacklisting, highlighting the difficulty of isolating the causal role of PMV from broader policy enforcement. Secondary forest regrowth effects remain more robust, suggesting PMV's potential role in promoting reforestation when paired with local governance support.

The total net present value (NPV) of benefits accrued over a 10-year period amounts to approximately $6.98 billion (2020 USD, full sample) or $0.38 billion (2020 USD, restricted sample), based on a social cost of carbon (SCC) of $50, which significantly outweighs the program's initial cost of $12.1 million. These figures indicate that the PMV program is a cost-effective policy that has yielded substantial environmental benefits.

Our research reveals that a substantial portion of these positive outcomes stems from rural properties registered under the Brazilian Rural Environmental Registry (CAR), predominantly from medium-sized properties. Such findings resonate with our initial hypotheses, highlighting the intertwined relationship between improved land regularization, tenure security, and beneficial land use outcomes. Beyond the realm of environmental impact, the PMV has also proven instrumental in mitigating land-related conflicts and bolstering non-agricultural economic activity within its participant municipalities.

In conclusion, our paper underscores the potential of voluntary programs that prioritize local governance enhancement. Policies like PMV can lead to broader positive benefits that transcend their initial targets. With the global urgency to address environmental degradation, understanding and harnessing such beneficial effects could pave the way for more holistic and effective environmental interventions. The success of the PMV in promoting secondary forest growth exemplifies the critical importance of considering all environmental outcomes when evaluating land use policies. However, these types of policies might need

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to go hand in hand with stricter command and control policies like the blacklisting of municipalities in order to be effective. Decentralizing environmental governance and making local governments more effective remains a promising avenue for achieving sustainable environmental management, albeit within a context of strong federal institutions and strict command-and-control.

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During the preparation of this work the authors used ChatGPT in order to edit the grammar of the writing. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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A Description of the Data

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Table A.1: Description of the Data

	Description of Variable	Source of Data
Municipality Characteristics		
Population	Number of inhabitants (N)	Population Estimates, Instituto Brasileiro de Geografia e Estatística - IBGE
Population Density	Number of inhabitants/ Municipality area (N/SqKm)	Population Estimates, and Municipality Shapefile from IBGE
GDP Capita	Gross domestic product per capita	SIDRA & IBGE
GDP	Gross domestic product (2019R$)	SIDRA
Rainfall	Annual avg. rainfall (mm)	TerraClimate
Temperature	Annual avg. temperature (K)	TerraClimate
Protected Areas	Extent of Protected Area (SqKm)	TerraBrasilis
Federal Roads	Extent of Federal Road Length (Km)	TerraBrasilis
Treatment Groups		
PMV	Dummy for municipality's year participation in PMV	P.M.V. 2021a; P.M.V. 2021b
PMVyear	Year of municipality's participation in PMV	"
Blacklisted	Dummy for municipality's year participation in MPs	MMA 2024
Blacklistedyear	Year of municipality's blacklisted by MPs	"
Outcomes		
Secondary Forest Growth	Annual extent of Secondary Forest (SqKm)	Junior et al. 2020 dataset
Young Secondary Forest Growth	Annual extent of Secondary Forests Age 1-5 (SqKm)	"
Middle Secondary Forest Growth	Annual extent of Secondary Forests Age 6-15 (SqKm)	"
Old Secondary Forest Growth	Annual extent of Secondary Forests Age 16-34 (SqKm)	"
Secondary Forest CAR	Annual extent of Secondary Forest CAR landholdings (SqKm)	Junior et al. 2020 dataset with Sistema de Cadastro Ambiental Rural (SICAR) 2020
Secondary Forest CAR Small		"
Secondary Forest CAR Medium		"
Secondary Forest CAR Large		"
Mapbiomas Deforestation	Annual Deforestaion from MapBiomas (SqKm)	MapBiomas Collection 8.1
PRODES Deforestation	Annual Deforestaion from PRODES (SqKm)	TerraBrasilis
Fines	Total Amount of Enviromental Fines (2019R$)	Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis - IBAMA
Fires	Annual sum of Fires in municipality (N)	Fire Information for Resource Management System - FIRMS
Land Conflict	Anuual sum of Land Conflicts in municipality (N)	Comissão Pastoral da Terra - CPT
Non Ag Value	Non agriculture gross value added (GVA) (2019R$)	Non Agricultural GVA (2019R$) from SIDRA

B The Importance of Secondary Forests: An Often Overlooked Outcome

Even with Brazil's lauded deforestation control policy, native forest cover has decreased significantly in the past few decades. However, as illustrated in Figure B.2, Brazil has also experienced an increase in areas covered by secondary forests and forest plantations (silviculture) between 1987 and 2019. Secondary forest recovery typically occurs in areas that were once clear-cut or degraded native forests, suggesting a spatial correlation between forest recovery and previous deforestation (Chokkalingam et al. 2001). In this sense, secondary forests can only occur in places that have been previously deforested, and as such the negative correlation is not surprising. However, the increase in secondary forests suggests that some of the land which is cleared is later allowed to recover and forests allowed to regrow. Understanding when and why this happens is the subject of a growing research agenda (Baragwanath et al. 2023; Camara et al. 2023; Heinrich et al. 2023)

The regeneration of secondary forests can play a pivotal role in successful, enduring climate policy. Notably, secondary forests in neotropical regions have been found to be highly efficient at carbon sequestration, with an estimated mean rate of 3.05Mg C $ha^{-1}yr^{-1}$ (Poorter et al. 2016). The growth of secondary forests can also counteract biodiversity loss (Lennox et al. 2018) and serve as a habitat for endangered and vulnerable species. Considering the benefits offered by secondary forest regeneration (Chazdon 2008; Chazdon et al. 2016a; Chazdon et al. 2016b; Matos et al. 2020), understanding the circumstances that promote secondary forest growth and developing policies that drive the successful restoration of native forests can be an important climate policy agenda.

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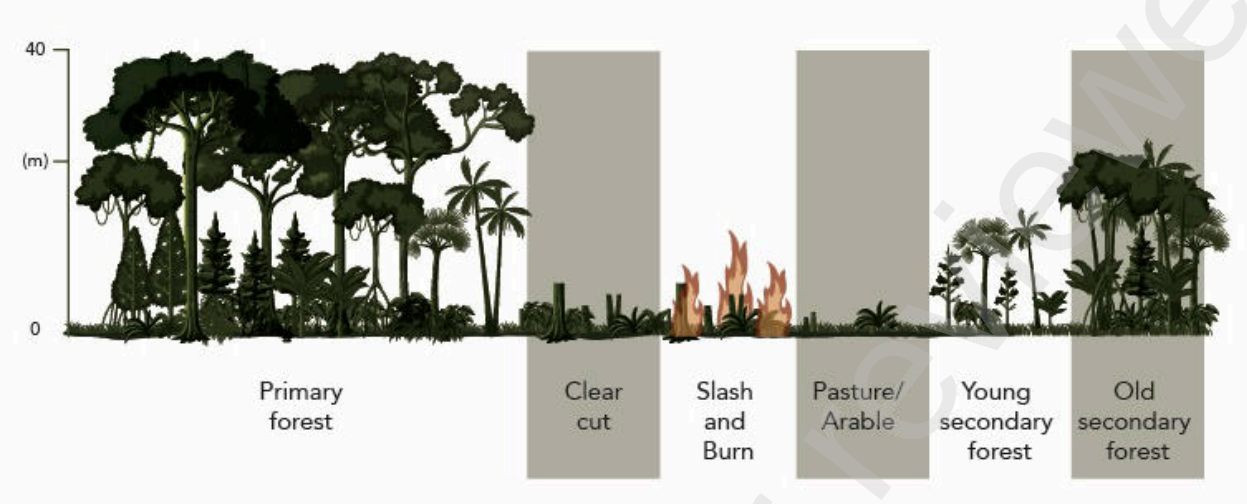
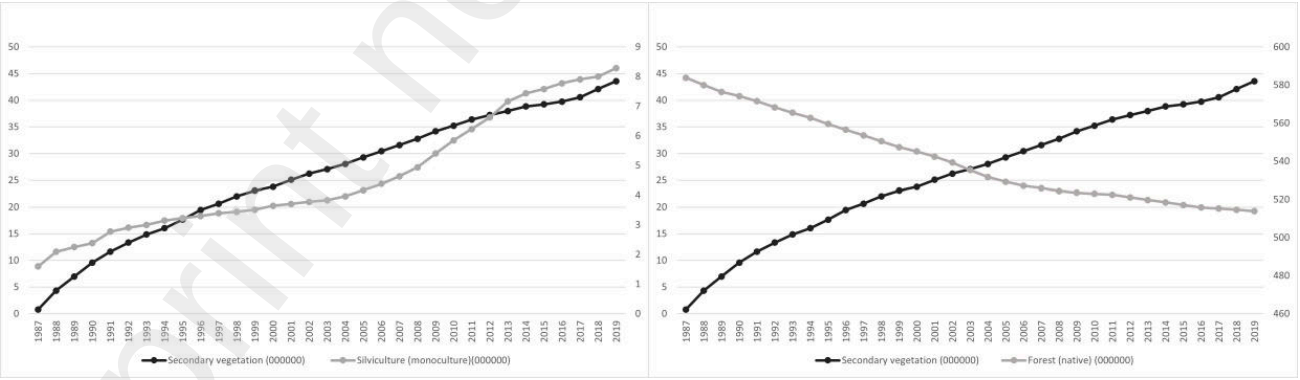


Figure B.1: Forest Transitions Cycle

Figure B.2: Forest transitions with respect to Secondary forest

(a) Silviculture (b) Native forest



Notes: Figure shows the area under secondary forest (on left y-axis), native forest, and forest plantations (both on right y-axis) in million hectares (1 km^2 = 100 hectares) between 1987 and 2019. We observe that both secondary forests and forest plantations expanded over this period; however, the area of secondary forest grew approximately 5.5 times more than that of forest plantations.

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C Maps

C.1 Blacklisting Enrollment

C.2 Maps of Secondary Forest Growth

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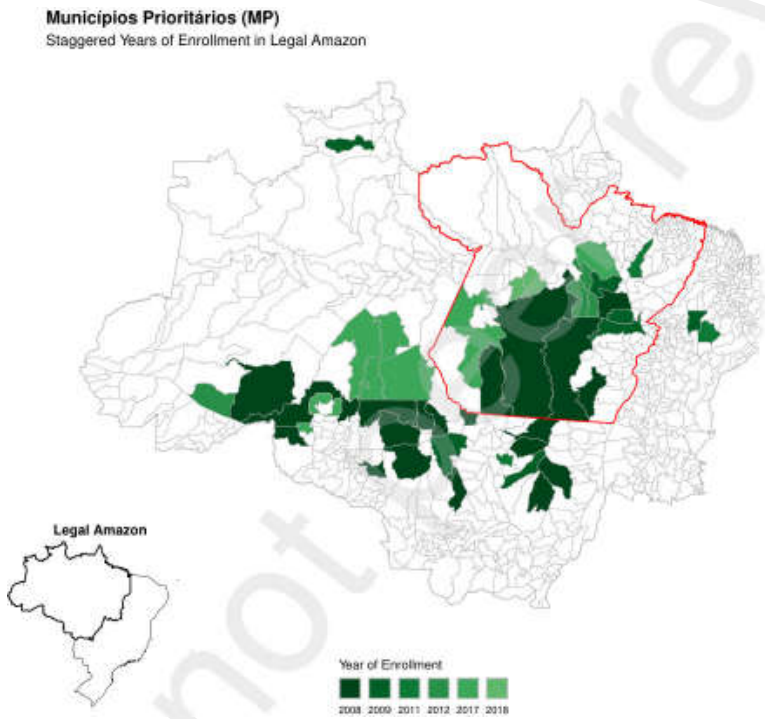


Figure C.3: Timing of Blacklisting of Brazilian Municipalities from 2008 to 2018. Staggered treatment of municipalities is shown where lighter shades of blue represent earlier adopters. The state of Pará is highlighted with a red border.

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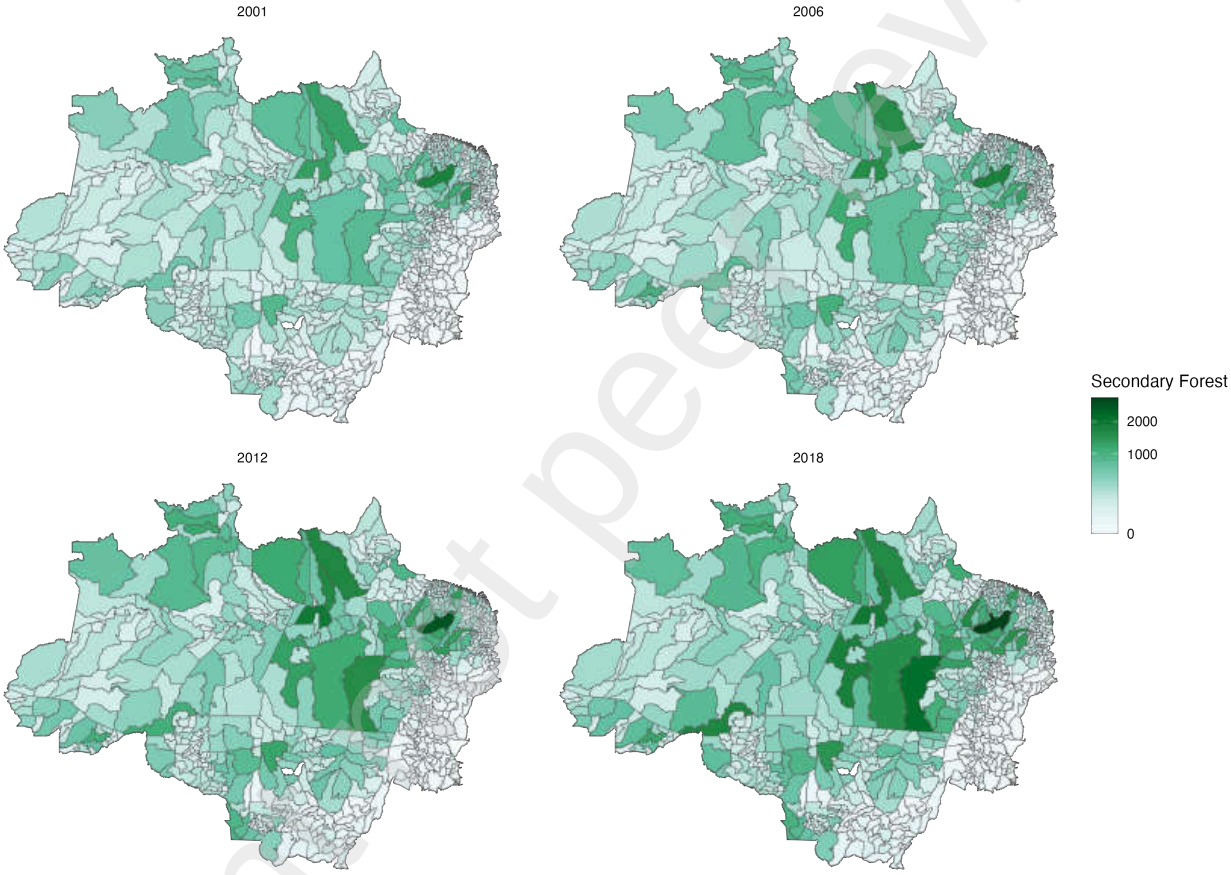


Figure C.4: Maps of Extent Secondary Forest Growth (km^2)

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C.3 Map of Deforestation

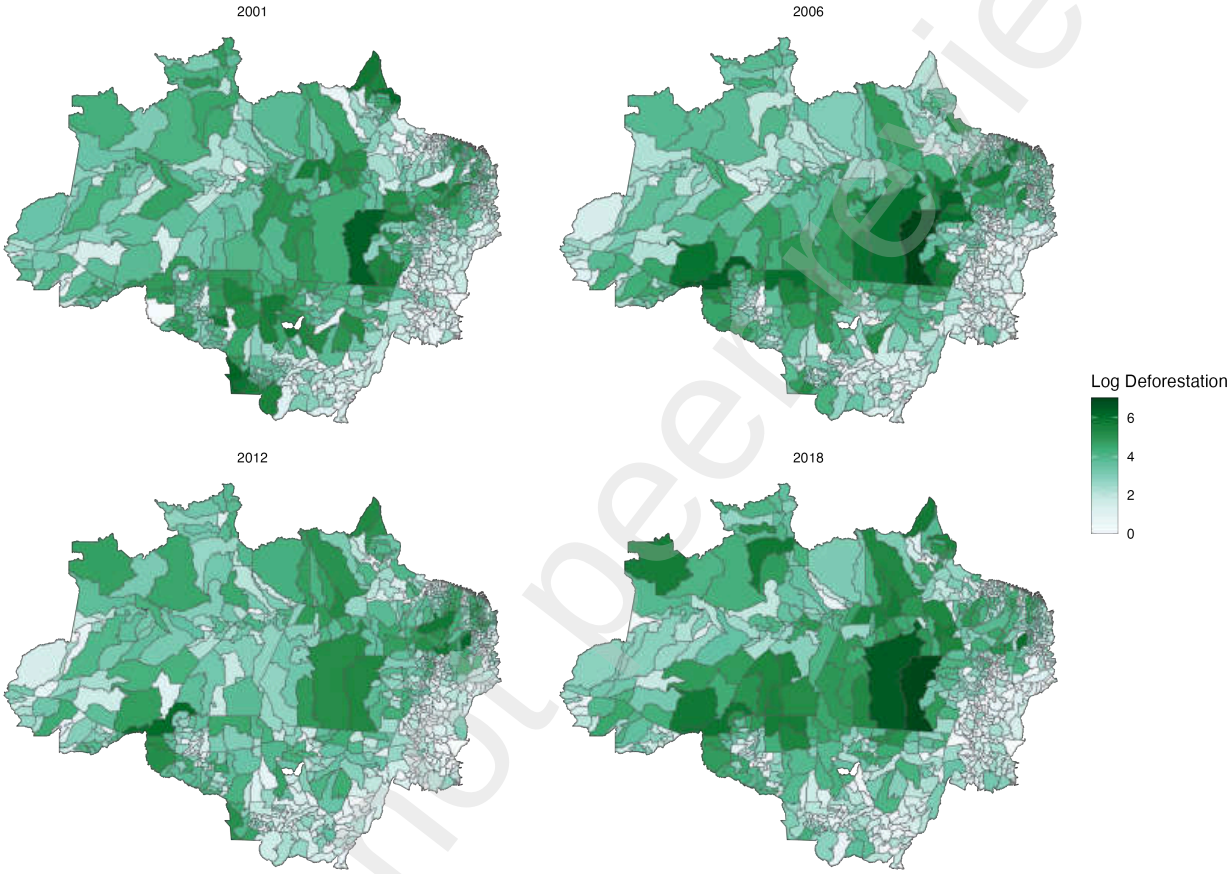


Figure C.5: Maps of Deforestation

D Description of Dependent Variables

D.1 Histograms of Secondary Forests By PMV Participation

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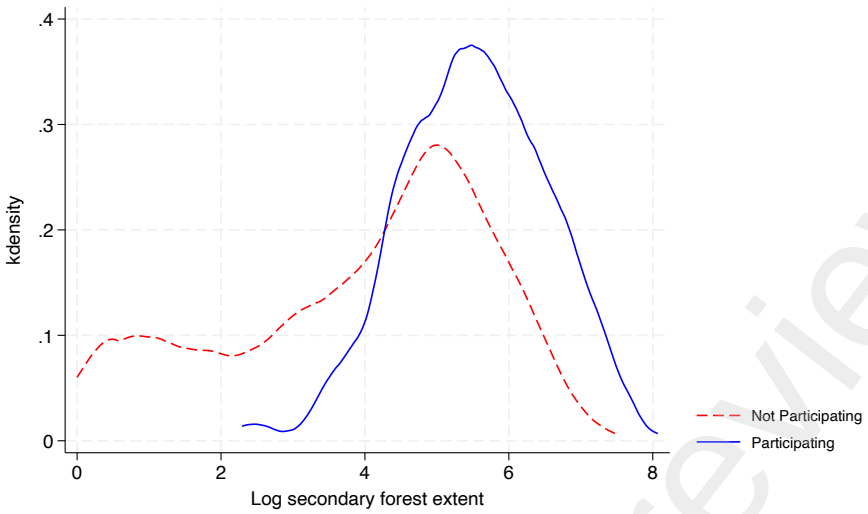


Figure D.6: Histogram of Log Secondary Forest

D.2 Trends in Secondary Forest Extents by PMV Participation

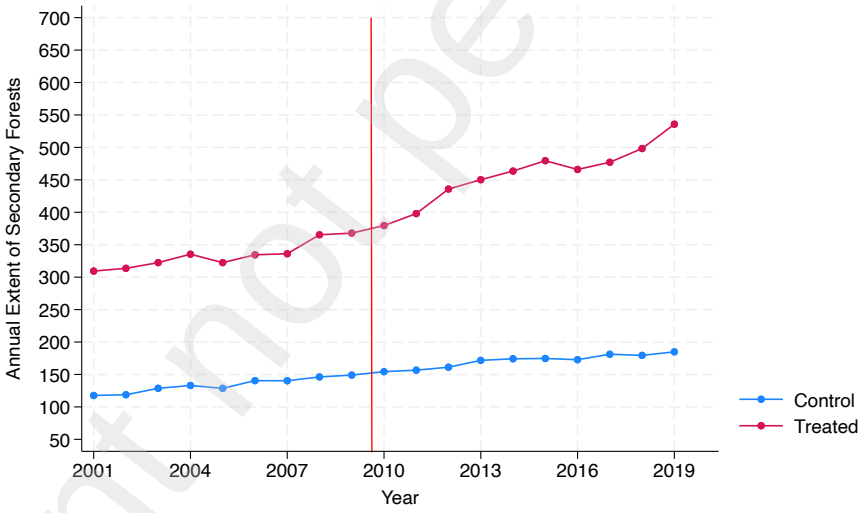


Figure D.7: Trends in Secondary Forest Extents by PMV Participation

D.3 Histogram of Young Secondary Forest Growth

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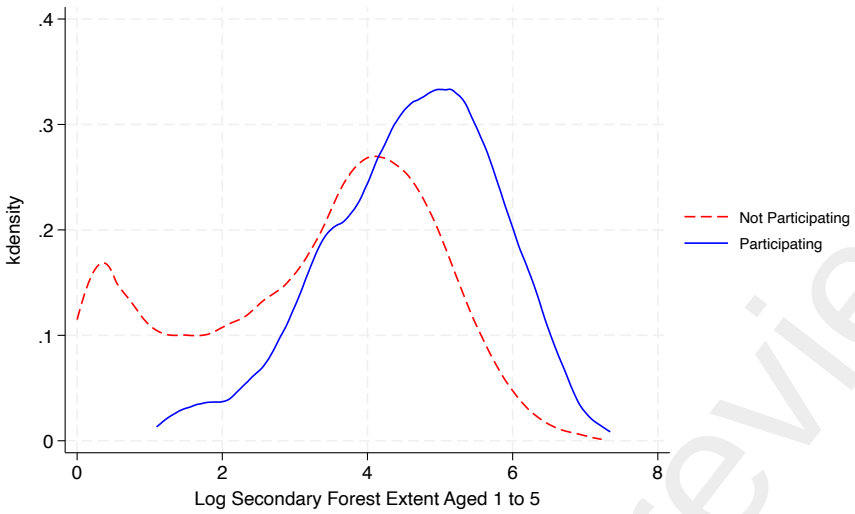


Figure D.8: Histogram of Young Secondary Forest

D.4 Histogram of Middle Secondary Forest Growth

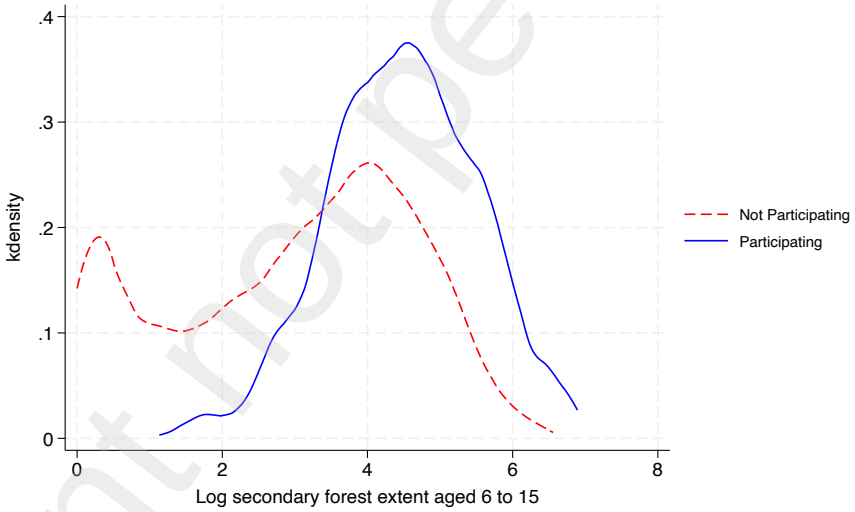


Figure D.9: Histogram of Middle Secondary Forest

D.5 Histogram of Old Secondary Forest Growth

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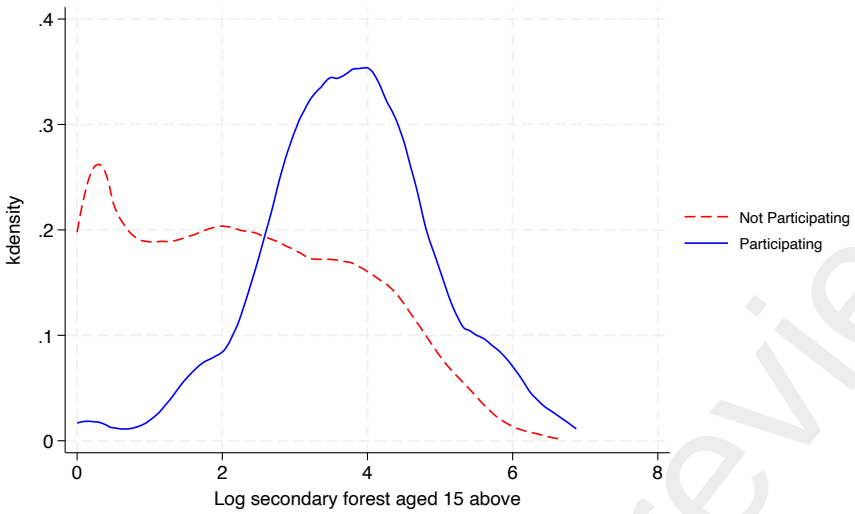


Figure D.10: Histogram of Old Secondary Forest

E Politics in Pará

Between 2009 and 2019, the state of Pará in Brazil had two significant political figures serving as governors: Ana Júlia Carepa and Simão Jatene. Here's an in-depth look at these governors, with a focus on their partisanship and their approach to environmental issues:

Ana Júlia Carepa (2007-2011)

- **Partisanship**: Ana Júlia Carepa was a member of the Workers' Party (PT), a left-wing political party in Brazil. Historically, the PT has been known for championing social causes, especially workers' rights, and has played a pivotal role in Brazil's political landscape.

- **Environmental Politics**: Carepa's tenure was marked by efforts to curtail deforestation in the Amazon region, which encompasses Pará. Despite facing challenges such as the vast size of the state, influential farming and logging lobbies, and the intricacies of implementing environmental regulations in distant locales, her administration initiated measures to boost satellite surveillance to combat illegal logging. Towards the end of her tenure in 2010, the "Programa Municípios Verdes" (Green Municipalities Program) was launched. This program aimed to combat deforestation by promoting sustainable practices, engaging local governments, and incentivizing municipalities to achieve and maintain a low deforestation rate. However, her tenure also saw tensions between environmentalists and local businesses, especially regarding land use and indigenous rights.

Simão Jatene (2011-2019)

- **Partisanship**: Simão Jatene represented the Brazilian Social Democracy Party (PSDB), a center-right political party in Brazil.

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- **Environmental Politics**: Simão Jatene's tenure also grappled with the challenges of deforestation and the need for sustainable development. He emphasized the importance of economic development, which sometimes led to clashes with environmentalists. However, during Jatene's tenure, the continuation and expansion of the "Programa Municípios Verdes" was evident. Jatene also worked on strengthening the state's environmental agencies and collaborating with international partners on conservation efforts.

F PMV Objectives

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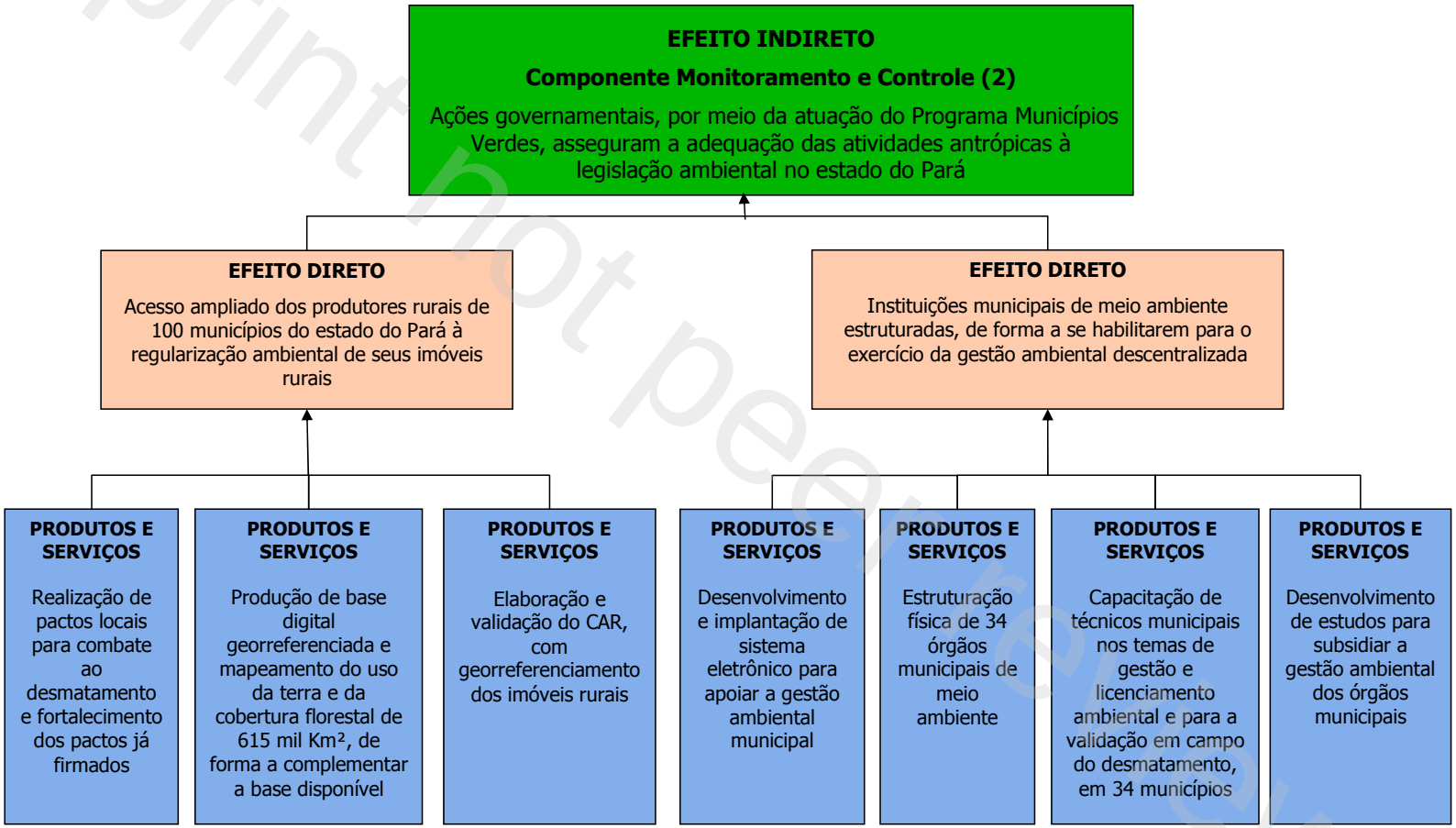
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Projeto: Programa Municípios Verdes

Responsável pelo projeto: Estado do Pará



G Robustness

G.1 Additional Results: Deforestation & Secondary Forests

G.1.1 Effects of PMV on Deforestation (Alternative Methods)

For robustness, we also estimate the effects using alternative methods: Two-Way Fixed Effects (TWFE), Callaway and Sant'Anna Difference-in-Differences (CSDID) (Callaway et al. 2021), and the Interactive Fixed Effects counterfactual model (IFEct) (Xu 2022). While all methods confirm a negative impact of PMV on deforestation, GSC-EM provides the most reliable estimates due to its ability to address pre-treatment differences and staggered treatment adoption more effectively.

Table G.2: Effects of PMV on Deforestation

	TWFE	CSDID	IFEct	GSC-EM
ATT PRODES	-0.31***	-0.31***	-0.44***	-0.44***
	(0.07)	(0.07)	(0.05)	(0.03)
ATT MapBiomas	-0.22***	-0.22***	-0.12***	-0.12***
	(0.09)	(0.09)	(0.06)	(0.04)
Covariates	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	12880	12880	12880	12880
Treated Munics	124	124	124	124
Control Munics	631	631	634	634
N Munics	755	755	758	758
Periods	17	17	17	17
Parametric	Yes	Yes	Yes	No

Notes: Table shows results from estimating 4 different causal panel models. The dependent variable for all models is the log rate of deforestation, and for all methods we include municipality and year fixed effects. Columns (1) and (2) display the average treatment effect on the treated estimated using CSDID (Callaway et al. 2021) both without (1) and with (2) covariates. Column (3) displays the results from the interactive fixed effects model (Liu et al. 2021), while column (4) shows the results from the Generalized Synthetic Controls Method using the EM method (Xu 2017a). Covariates included are mean precipitation, mean temperature, GDP per capita, population density and proportion of land covered by Protected Areas.
Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

The estimates from TWFE, CSDID, and IFEct confirm that PMV reduces deforestation rates. However, these methods assume parallel trends or rely on fixed effects, which may not fully account for selection into treatment and pre-existing differences between treated and control municipalities. As discussed in the main text, GSC-EM overcomes these limitations by explicitly modeling differential pre-treatment trends and structural differences, making it the most credible estimator for this analysis.

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G.1.2 Effects of PMV on Secondary Forests (Alternative Methods)

The estimates from CSDID and IFEct confirm that PMV increases secondary forest cover. However, these methods assume parallel trends or rely on fixed effects, which may not fully account for selection into treatment and pre-existing differences between treated and control municipalities. As discussed in the main text, GSC-EM overcomes these limitations by explicitly modeling differential pre-treatment trends and structural differences, making it the most credible estimator for this analysis.

Table G.3: Effects of PMV on Secondary Forests

	(1) CSDID	(2) CSDID Covs	(3) IFE	(4) GSC-EM
ATT Secondary Forest	0.09***	0.09***	0.06***	0.07***
	(0.02)	(0.02)	(0.02)	(0.00)
Covariates	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	12880	12880	12880	12880
Treated Munics	124	124	124	124
Control Munics	631	631	634	634
N Munics	755	755	758	758
Periods	17	17	17	17
Parametric	Yes	Yes	Yes	No

Notes: Table shows results from estimating 4 different causal panel models. The dependent variable for all models is the log of secondary forest extent, and for all methods we include municipality and year fixed effects. Columns (1) and (2) display the average treatment effect on the treated estimated using CSDID (Callaway et al. 2021) both without (1) and with (2) covariates. Column (3) displays the results from the interactive fixed effects model (Liu et al. 2021), while column (4) shows the results from the Generalized Synthetic Controls Method using the EM method (Xu 2017a). Covariates included are mean precipitation, mean temperature, GDP per capita, population density and proportion of land covered by Protected Areas.
Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

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Table G.4: Effects of PMV on Secondary Forests by Age of Secondary Forest

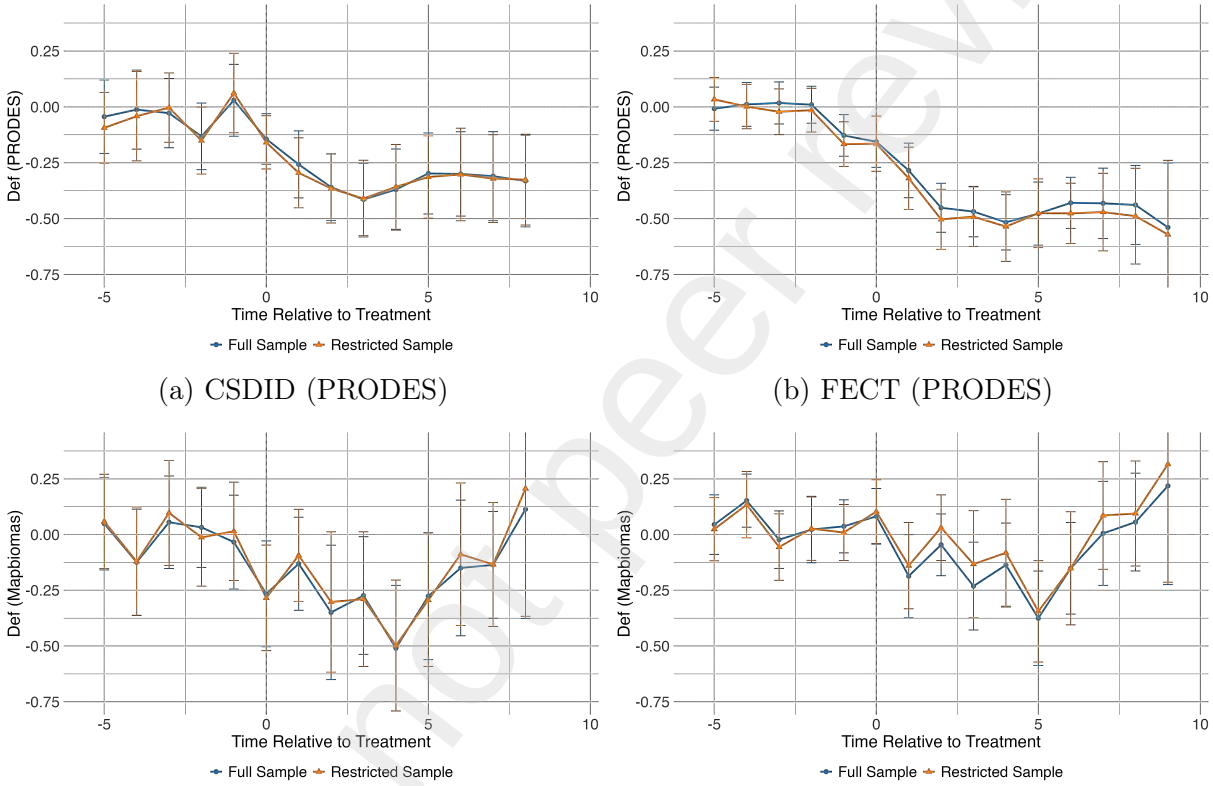
	(1) CSDID	(2) CSDID Covs	(3) IFE	(4) Gsynth EM
ATT Young Forest	0.13***	0.15***	0.10***	0.11***
	(0.03)	(0.04)	(0.03)	(0.01)
ATT Middle Forest	-0.01	-0.01	0.00	0.01*
	(0.04)	(0.04)	(0.04)	(0.00)
ATT Old Forest	0.06	0.07	0.03	0.03*
	(0.04)	(0.04)	(0.05)	(0.01)
Covariates	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	12880	12880	12880	12880
Treated Munics	124	124	124	124
Control Munics	631	631	634	634
N Munics	755	755	758	758
Periods	17	17	17	17
Parametric	Yes	Yes	Yes	No

Notes: Table shows results from estimating 4 different causal panel models. The results in the first row of the table are for the dependent variable logarithm of young secondary forest extents, followed by the results for the dependent variable logarithm of middle-aged secondary forest extents, and finally the results for the dependent variable logarithm of old secondary forest extents. For all methods, we include municipality and year fixed effects. Columns (1) and (2) display the average treatment effect on the treated estimated using CSDID (Callaway et al. 2021) both without (1) and with (2) covariates. Column (3) displays the results from the interactive fixed effects model (Liu et al. 2021), while column (4) shows the results from the Generalized Synthetic Controls Method using the EM method (Xu 2017a). Covariates included are mean precipitation, mean temperature, GDP per capita, population density and proportion of land covered by Protected Areas.
Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

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Figure G.11: Effects of PMV on Deforestation (Mapbiomas and PRODES)



(a) CSDID (PRODES)

(b) FECT (PRODES)

(c) CSDID (Mapbiomas)

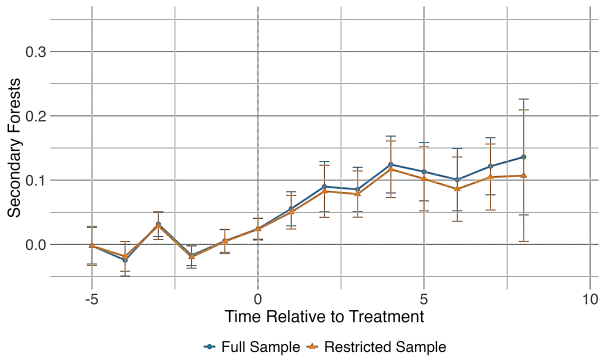
(d) FECT (Mapbiomas)

Notes: This figure compares estimates for log deforestation using CSDID and FECT models, for both PRODES (top row) and Mapbiomas (bottom row) as outcome variables. Each panel shows results for the full sample and the sample excluding blacklisted municipalities. The CSDID plots use a staggered DiD estimator, and the FECT plots use an interactive fixed effects model. Covariates include precipitation, temperature, GDP per capita, population density, and protected areas coverage.

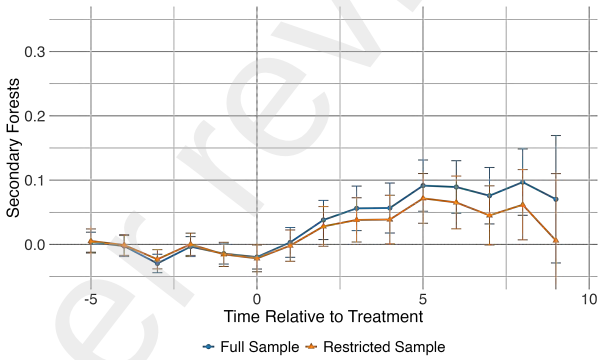
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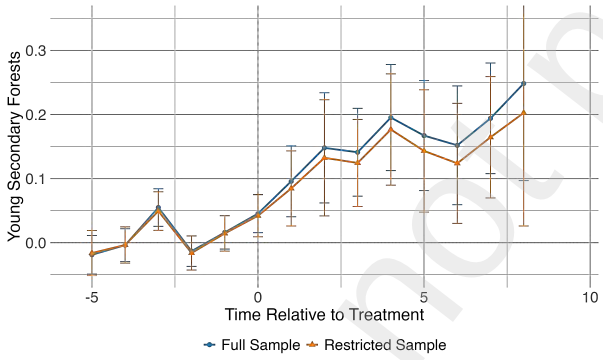
Figure G.12: Effects of PMV on Secondary Forests (All and Young)



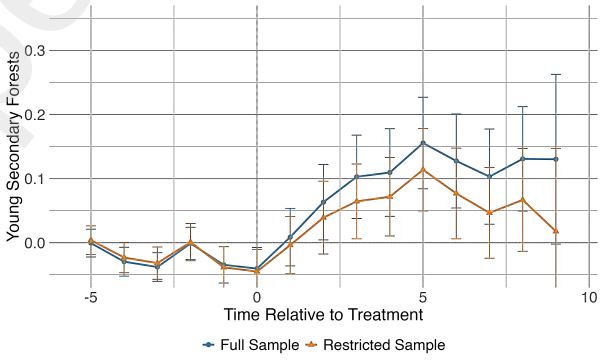
(a) CSDID Secondary Forests



(b) FECT Secondary Forests



(c) CSDID Young Secondary Forests



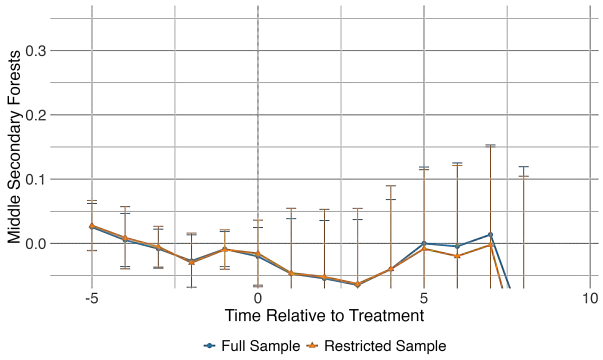
(d) FECT Young Secondary Forests

Notes: This figure presents results for secondary forest outcomes. Panels (a) and (b) show results using the overall secondary forest share as the dependent variable, while panels (c) and (d) focus on young secondary forests (1–8 years). We compare full sample estimates with those excluding blacklisted municipalities. Covariates include precipitation, temperature, GDP per capita, population density, and protected areas coverage.

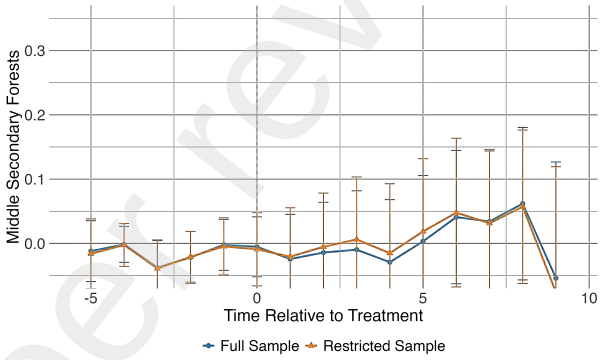
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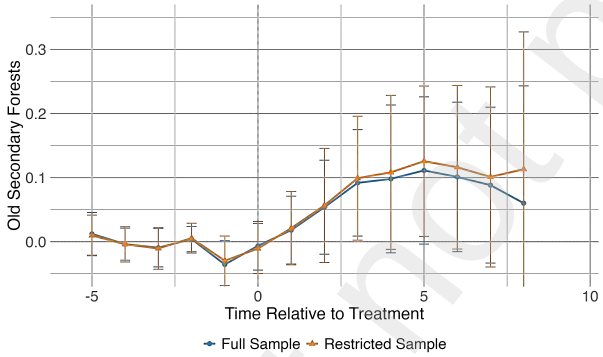
Figure G.13: Effects of PMV on Secondary Forests (Middle and Old)



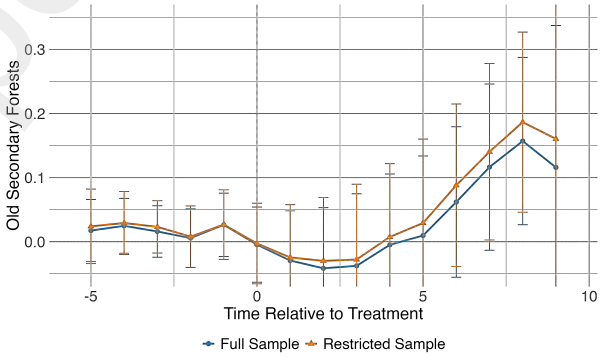
(a) CSDID Middle Secondary Forests



(b) FECT Middle Secondary Forests



(c) CSDID Old Secondary Forests



(d) FECT Old Secondary Forests

Notes: Panels (a) and (b) show results for middle-aged secondary forests (9–16 years), while panels (c) and (d) show results for older secondary forests (17–24 years). All plots compare the full sample with the restricted sample. Covariates include precipitation, temperature, GDP per capita, population density, and protected areas coverage.

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G.1.3 Generalized Synthetic Controls Counterfactuals

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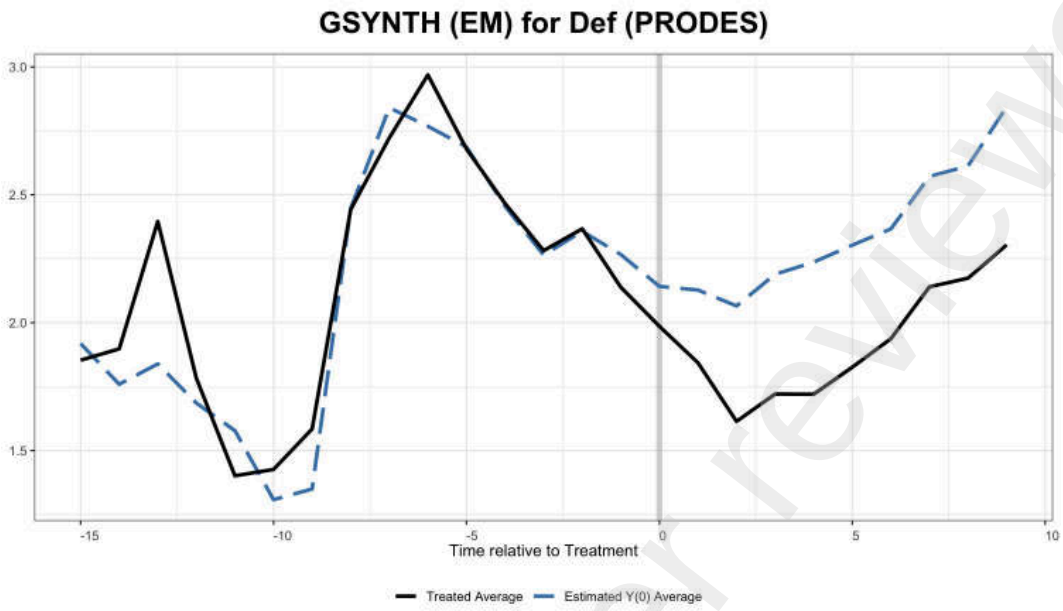


Figure G.14: GSYNTH Counterfactuals: PRODES Deforestation

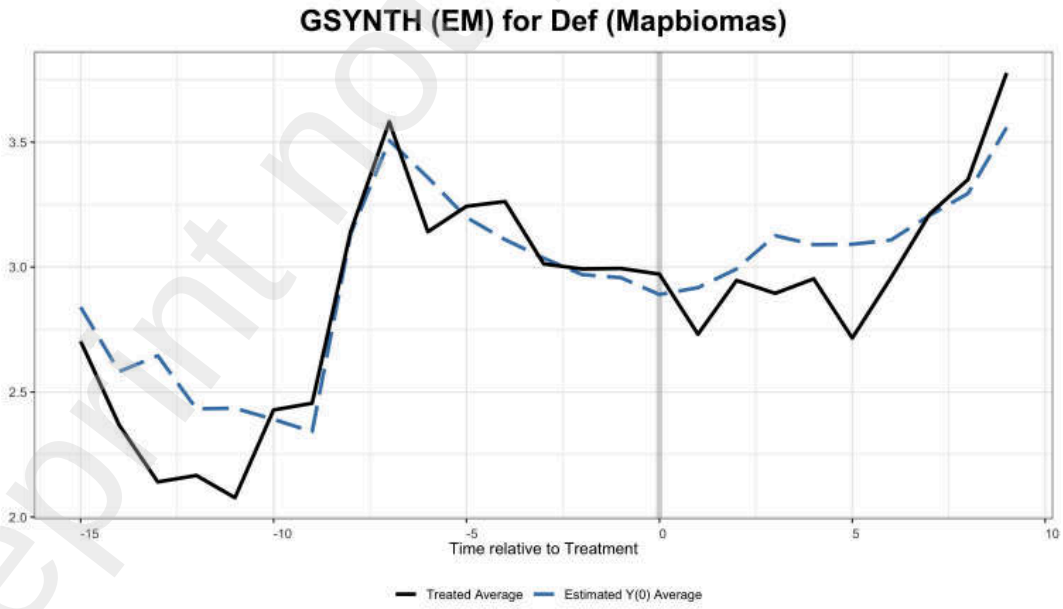


Figure G.15: GSYNTH Counterfactuals: Mapbiomas Deforestation

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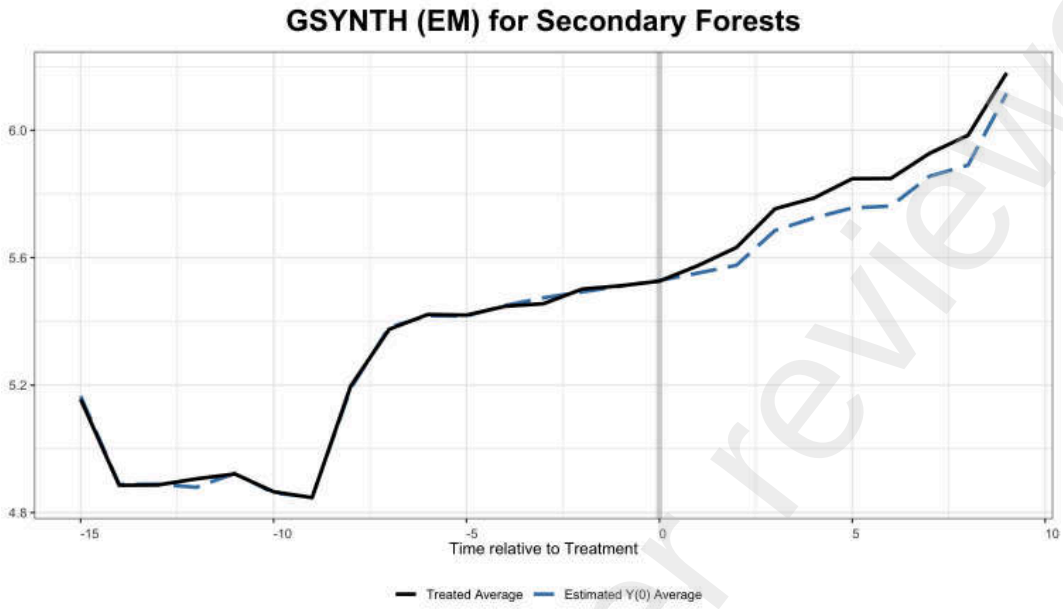


Figure G.16: GSYNTH Counterfactuals: Secondary Forests

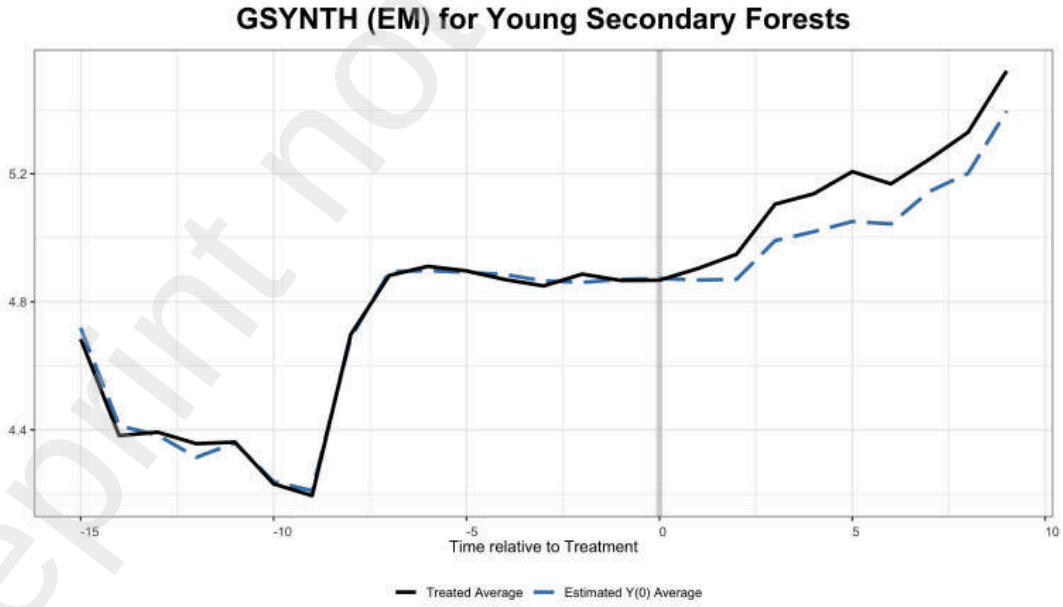


Figure G.17: GSYNTH Counterfactuals: Young Secondary Forests

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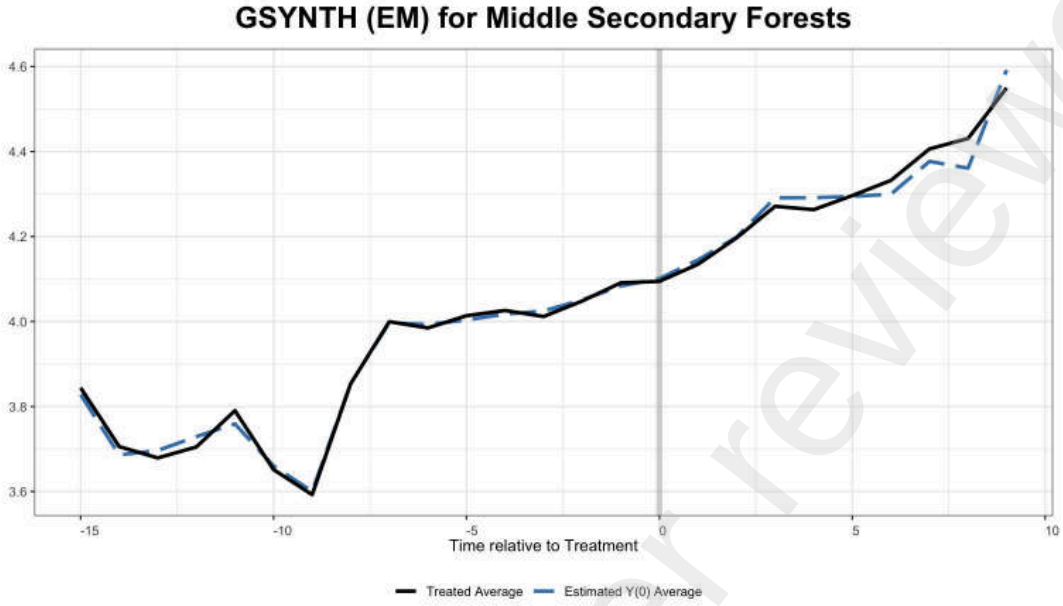


Figure G.18: GSYNTH Counterfactuals: Middle Secondary Forests

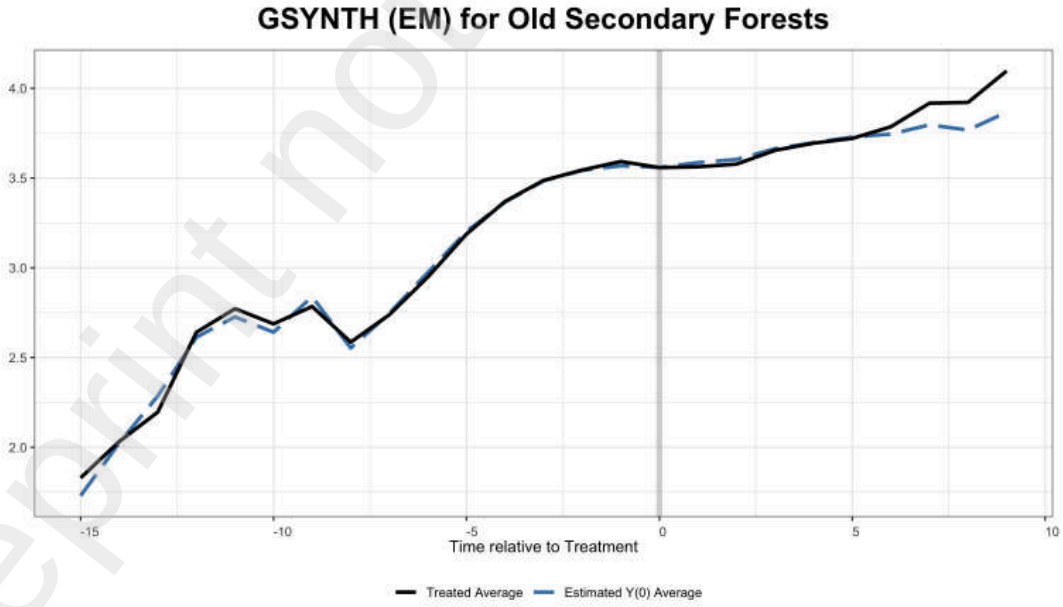


Figure G.19: GSYNTH Counterfactuals: Old Secondary Forests

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H Additional Results: Mechanisms

Table H.5: Effects of PMV on Deforestation Within CAR Properties, By Property Size

	(1) CSDID	(2) CSDID Covs	(3) IFE	(4) Gsynth EM
ATT Deforestation CAR, All	-0.01	0.01	-0.03***	-0.03***
	(0.00)	(0.00)	(0.00)	(0.00)
ATT Deforestation CAR, Small	-0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)
ATT Deforestation CAR, Medium	-0.01	-0.01**	-0.02**	-0.02***
	(0.01)	(0.00)	(0.01)	(0.00)
ATT Deforestation CAR, Large	0.01	0.01	-0.01	-0.01
	(0.01)	(0.01)	(0.02)	(0.01)
Covariates	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Treated Munics	124	124	123	123
Control Munics	631	631	618	618
Periods	17	17	17	17
Parametric	Yes	Yes	Yes	No

Notes: Table shows results from estimating 4 different causal panel models. For all methods, we include municipality and year fixed effects. Columns (1) and (2) display the average treatment effect on the treated estimated using CSDID (Callaway et al. 2021) both without (1) and with (2) covariates. Column (3) displays the results from the interactive fixed effects model (Liu et al. 2021), while column (4) shows the results from the Generalized Synthetic Controls Method using the EM method (Xu 2017a). Covariates included are mean precipitation, mean temperature, GDP per capita, population density and proportion of land covered by Protected Areas.
Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

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Table H.6: Effects of PMV on Secondary Forests Within CAR Properties, By Property Size

	(1) CSDID	(2) CSDID Covs	(3) IFE	(4) Gsynth EM
ATT Secondary Forest CAR, All	-0.12	0.05	0.40***	0.40***
	(0.12)	(0.11)	(0.09)	(0.07)
ATT Secondary Forest CAR, Small	-0.05	0.01	0.04	0.04
	(0.06)	(0.07)	(0.03)	(0.04)
ATT Secondary Forest CAR, Medium	-0.07	0.02	0.26***	0.23***
	(0.06)	(0.07)	(0.06)	(0.03)
ATT Secondary Forest CAR, Large	-0.00	0.02	0.10***	0.07***
	(0.02)	(0.02)	(0.02)	(0.01)
Covariates	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Treated Munics	124	124	123	123
Control Munics	631	631	618	618
Periods	17	17	17	17
Parametric	Yes	Yes	Yes	No

Notes: Table shows results from estimating 4 different causal panel models. For all methods, we include municipality and year fixed effects. Columns (1) and (2) display the average treatment effect on the treated estimated using CSDID (Callaway et al. 2021) both without (1) and with (2) covariates. Column (3) displays the results from the interactive fixed effects model (Liu et al. 2021), while column (4) shows the results from the Generalized Synthetic Controls Method using the EM method (Xu 2017a). Covariates included are mean precipitation, mean temperature, GDP per capita, population density and proportion of land covered by Protected Areas.
Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

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Table H.7: Mechanisms: Effects of PMV on Fines, Fires and Land Conflict

	(1) CSDID	(2) CSDID Covs	(3) IFE	(4) Gsynth EM
ATT Fines	-0.73*	-0.85**	-0.69***	-0.69***
	(0.40)	(0.43)	(0.22)	(0.17)
ATT Fires	-5.59	-4.33	-31.53*	-31.98***
	(12.83)	(13.88)	(16.87)	(2.96)
ATT Land Conflict	-0.00	-0.00	-0.02***	-0.02***
	(0.01)	(0.01)	(0.00)	(0.00)
ATT Non Ag Value	0.06	0.05	0.05*	0.05***
	(0.04)	(0.04)	(0.03)	(0.00)
Covariates	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	12880	12880	12880	12880
Treated Munics	124	124	124	124
Control Munics	631	631	634	634
N Munics	755	755	758	758
Periods	17	17	17	17
Parametric	Yes	Yes	Yes	No

Notes: Table shows results from estimating 4 different causal panel models. For all methods, we include municipality and year fixed effects. Columns (1) and (2) display the average treatment effect on the treated estimated using CSDID (Callaway et al. 2021) both without (1) and with (2) covariates. Column (3) displays the results from the interactive fixed effects model (Liu et al. 2021), while column (4) shows the results from the Generalized Synthetic Controls Method using the EM method (Xu 2017a). Covariates included are mean precipitation, mean temperature, GDP per capita, population density and proportion of land covered by Protected Areas.
Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

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I Additional Results: Effects of PMV Excluding Ever Blacklisted Municipalities

Table I.8: Effects of PMV on Secondary Forests and Deforestation Excluding All Municipalities that were Ever Blacklisted

	(1) CSDID	(2) CSDID Covs	(3) IFE	(4) Gsynth EM
ATT Secondary Forest	0.08***	0.08***	0.04***	0.05***
	(0.02)	(0.02)	(0.02)	(0.00)
ATT Prodes Deforestation	-0.32***	-0.32***	-0.47***	-0.47***
	(0.07)	(0.07)	(0.07)	(0.03)
ATT Mapbiomas Deforestation	0.20*	-0.20*	-0.07	-0.07
	(0.11)	(0.10)	(0.07)	(0.05)
Covariates	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Treated Munics	124	124	111	111
Control Munics	631	631	604	604
Periods	17	17	17	17
Parametric	Yes	Yes	Yes	No

Notes: Table shows results from estimating 4 different causal panel models. For all methods, we include municipality and year fixed effects. Columns (1) and (2) display the average treatment effect on the treated estimated using CSDID (Callaway et al. 2021) both without (1) and with (2) covariates. Column (3) displays the results from the interactive fixed effects model (Liu et al. 2021), while column (4) shows the results from the Generalized Synthetic Controls Method using EM method (Xu 2017a). Covariates included are mean precipitation, mean temperature, GDP per capita, population density and proportion of land covered by Protected Areas.
Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

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Table I.9: Mechanisms: Effects of Blacklisting on Fines, Fires, Land Conflict and GVA-agriculture

	(1) CSDID	(2) CSDID Covs	(3) IFE	(4) Gsynth EM
ATT Fines	0.02	1.10	1.59***	2.57***
	(0.42)	(0.91)	(0.21)	(0.82)
ATT Fires	-377.94***	-97.14	-292.50***	-32.10***
	(62.93)	(73.30)	(46.78)	(5.97)
ATT Land Conflict	-0.00	0.04	-0.00	0.00
	(0.01)	(0.07)	(0.01)	(0.01)
ATT Non Ag Value	0.06	0.12	0.02	-0.05***
	(0.04)	(0.06)	(0.04)	(0.01)
Covariates	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N Munics	755	755	758	758
Periods	17	17	17	17
Parametric	Yes	Yes	Yes	No

Notes: Table shows results from estimating 4 different causal panel models. For all methods, we include municipality and year fixed effects. Columns (1) and (2) display the average treatment effect on the treated estimated using CSDID (Callaway et al. 2021) both without (1) and with (2) covariates. Column (3) displays the results from the interactive fixed effects model (Liu et al. 2021), while column (4) shows the results from the Generalized Synthetic Controls Method using the EM method (Xu 2017a). Covariates included are mean precipitation, mean temperature, GDP per capita, population density and proportion of land covered by Protected Areas.
Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

J Carbon removal and avoided emission

Our main objective is to estimate the economic benefits of reducing deforestation and promoting secondary forest recovery under the Green Municipality Program (PMV) in the state of Pará, Brazil. We do this by quantifying the changes in CO_2 emissions—specifically, avoided emissions from reduced deforestation and additional carbon removals from forest regrowth—and valuing these changes using a range of social cost of carbon (SCC) estimates.

We use a panel of 144 municipalities in Pará from 2001 to 2019. Land use, emissions, and removals data are compiled from MapBiomas (Mapbiomas 2020), PRODES, and SEEG (De Azevedo et al. 2018). We estimate the average treatment effects (ATT) of PMV adoption on land use using a difference-in-differences framework. Based on our conservative ATT estimates, we assume a 12% reduction in deforestation and a 7% increase in secondary forest recovery attributable to PMV, starting in the year of adoption. These percentages are applied to observed land use in treated municipalities to calculate additional avoided deforestation and forest recovery over the 10-year post-treatment period (2010–2019).

Carbon impacts are derived by multiplying avoided deforestation and forest recovery by

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their respective emission and removal factors from SEEG. These yield total avoided CO_2 emissions (in million metric tons of carbon dioxide equivalent, $MtCO_2e$) and added CO_2 removals (also in $MtCO_2e$).

To monetize the carbon impacts, we apply a range of SCC values—50, 100, 150, 185, and 200 USD per $MtCO_2e$—including the updated estimate from Rennert et al. (2022). Each SCC value is multiplied by the corresponding emission reductions or removals to estimate total gross benefits. We then compute Net Present Values (NPVs) using a 5% discount rate and a base year of 2020, following standard practices in environmental economics.

We present results for two samples in Table J.10:

- **Full sample:** Includes all treated and control municipalities, including those ever blacklisted by IBAMA.
- **Restricted sample:** Excludes all municipalities that were ever blacklisted.

For each sample, we report the total area of avoided deforestation and secondary forest recovery (in km^2), along with the resulting carbon outcomes (in $MtCO_2e$) and associated economic values (in 2019 USD billions). These are summarized under multiple SCC scenarios.

In the restricted sample, the estimated ATT for avoided deforestation is not statistically significant. Accordingly, we conservatively set the avoided deforestation impact to zero. This implies that both the associated CO_2 emissions avoided and their economic value (both gross and NPV) are also zero in this sample. However, we still detect statistically significant increases in secondary forest recovery, which generate measurable carbon removals and economic benefits.

All final estimates are aggregated across the post-treatment period, saved into structured CSV files, and used to produce Table J.10 and related figures.

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Table J.10: Benefits of PMV: Full vs. Restricted Sample

Panel A: Full Sample – All Metrics								
SCC ($/tCO$_2$e)	Avoided Emissions Benefit (USD)	Forest Recovery Benefit (USD)	Total Benefit (USD)	NPV Avoided Emissions (USD)	NPV Forest Recovery (USD)	Total NPV (USD)	Emissions Avoided (MtCO$_2$e)	Carbon Removals (MtCO$_2$e)
50	8.12	0.92	9.04	6.27	0.71	6.98	162.36	18.35
100	16.24	1.83	18.07	12.53	1.43	13.96	162.36	18.35
150	24.35	2.75	27.11	18.80	2.14	20.94	162.36	18.35
185	30.04	3.39	33.43	23.19	2.64	25.83	162.36	18.35
200	32.47	3.67	36.14	25.07	2.85	27.92	162.36	18.35
Area Metrics (Full Sample)								
Avoided Deforestation (km^2)	5,683.68							
Secondary Forest Recovery (km^2)	36,663.06							
Panel B: Restricted Sample – All Metrics								
SCC ($/tCO$_2$e)	Avoided Emissions Benefit (USD)	Forest Recovery Benefit (USD)	Total Benefit (USD)	NPV Avoided Emissions (USD)	NPV Forest Recovery (USD)	Total NPV (USD)	Emissions Avoided (MtCO$_2$e)	Carbon Removals (MtCO$_2$e)
50	0.00	0.49	0.49	0.00	0.38	0.38	0.00	9.81
100	0.00	0.98	0.98	0.00	0.76	0.76	0.00	9.81
150	0.00	1.47	1.47	0.00	1.14	1.14	0.00	9.81
185	0.00	1.81	1.81	0.00	1.41	1.41	0.00	9.81
200	0.00	1.96	1.96	0.00	1.52	1.52	0.00	9.81
Area Metrics (Restricted Sample)								
Avoided Deforestation (km^2)	0.00							
Secondary Forest Recovery (km^2)	19,620.39							

Notes: This table reports biophysical and economic outcomes under varying social cost of carbon (SCC) assumptions. Metrics are disaggregated by full and restricted samples. Carbon emissions and removals are measured in tons (MtCO$_2$e); monetary values are in billions of 2019 USD. NPV assumes a 5% discount rate v